

Rewiring the market time clock in Spanish electricity markets: strategic behaviour and market power with higher granularity

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Abstract

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Glossary of acronyms

Acronym	Meaning
CCGT	Combined-cycle gas turbine: a flexible, dispatchable thermal generator.
DA	Day-ahead wholesale electricity market, cleared once per day.
IDA	Intraday auctions: sequence of (European) auctions held after the day-ahead clearing.
MTU	Market time unit: size of the time interval at which bids are submitted and prices clear.
REE	Red Eléctrica de España, the Spanish transmission-system operator.
OMIE	Operador del Mercado Ibérico de Energía, the market operator for the Spanish and Portuguese wholesale electricity markets.
TTF	Title Transfer Facility, the Dutch virtual gas hub and reference price for European gas.

1 Introduction

1.1 Motivation and research question

1.2 Related literature

1.3 Mechanism summary and contribution

TWO DISTINCT PURPOSES OF MTU15:

- OPERATIONAL / SYSTEM-OPERATOR CHANNEL. LETTING THE SYSTEM FOLLOW WITHIN-HOUR SOLAR VARIABILITY (CLOUD COVER, RAMP). MIDDAY IS EXACTLY WHERE THIS MATTERS MOST.
- STRATEGIC-EXPLOITATION CHANNEL. LETTING PIVOTAL FIRMS RE-PRICE ACROSS QUARTERS WHEN NET-LOAD MOVES SYSTEMATICALLY WITHIN THE HOUR. THIS IS WHAT DOMINANT CCGT CAN MONETIZE. **THIS IS WHAT WE DO!!**

1.4 Roadmap

2 Theoretical model

2.1 Setup

2.2 Equilibrium under low granularity

2.3 Equilibrium under high granularity

2.4 Predictions: dominant vs fringe response in time-varying vs flat subperiods

TODO: COROLLARY ON RAMP-CONSTRAINED HOURS. WHEN THE SYSTEM OPERATOR IMPOSES BINDING RAMP RESTRICTIONS (QUANTITY CANNOT ADJUST MUCH ACROSS CONSECUTIVE QUARTERS), THE HIGH-GRANULARITY REGIME FORCES FIRMS TO AMPLIFY THEIR STRATEGIC RESPONSE ON THE PRICE MARGIN RATHER THAN THE QUANTITY MARGIN. THE MTU15 REFORM THEREFORE AMPLIFIES MARKET POWER IN PRECISELY THOSE HOURS WHERE PHYSICAL ADJUSTMENT IS CONSTRAINED.

3 Institutional setting, data, and identification

3.1 Spanish wholesale electricity market: a sequential-market overview

STYLISTED MARKET DIAGRAM TBD HERE AS TIKZ: DA - IDA AUCTIONS - CONTINUOUS - BALANCING

3.2 The MTU15 reform sequence

3.3 Data sources

OMIE

- Day-ahead data:

- Intraday-auction data:
- Continuous intraday market data:

ESIOS

ENTSO-E

INCLUDE SAMPLE SIZE AND FAMILIES.

3.4 Critical-vs-flat hours: empirical calibration

Hours are classified by the **within-hour variance of residual demand**,

$$\sigma_{\text{within}}^2(h) = \mathbb{E}_d \left[\frac{1}{4} \sum_{q=1}^4 (D_{h,q,d} - \bar{D}_{h,d})^2 \right], \quad D_{h,q,d} = \text{demand}_{h,q,d} - \text{wind}_{h,q,d} - \text{solar}_{h,q,d}. \quad (1)$$

Critical hours: high σ_{within}^2 , 05:00–09:00 and 16:00–23:00 (eleven clock-hours). **Flat hours:** low σ_{within}^2 , 01:00–04:00 (three clock-hours). Borderline hours are dropped. Full 24-hour ranking in Appendix B.6.

OJO, IF WE INTRODUCE THE 11:00-15:00 PERIOD IN THE CONTROL, WE ARE MIXING THE OPERATIONAL AND STRATEGIC CHANNELS OF THE REFORM, BECAUSE THE OPERATIONAL EFFECT IS CONCENTRATED IN THOSE HOURS!!!

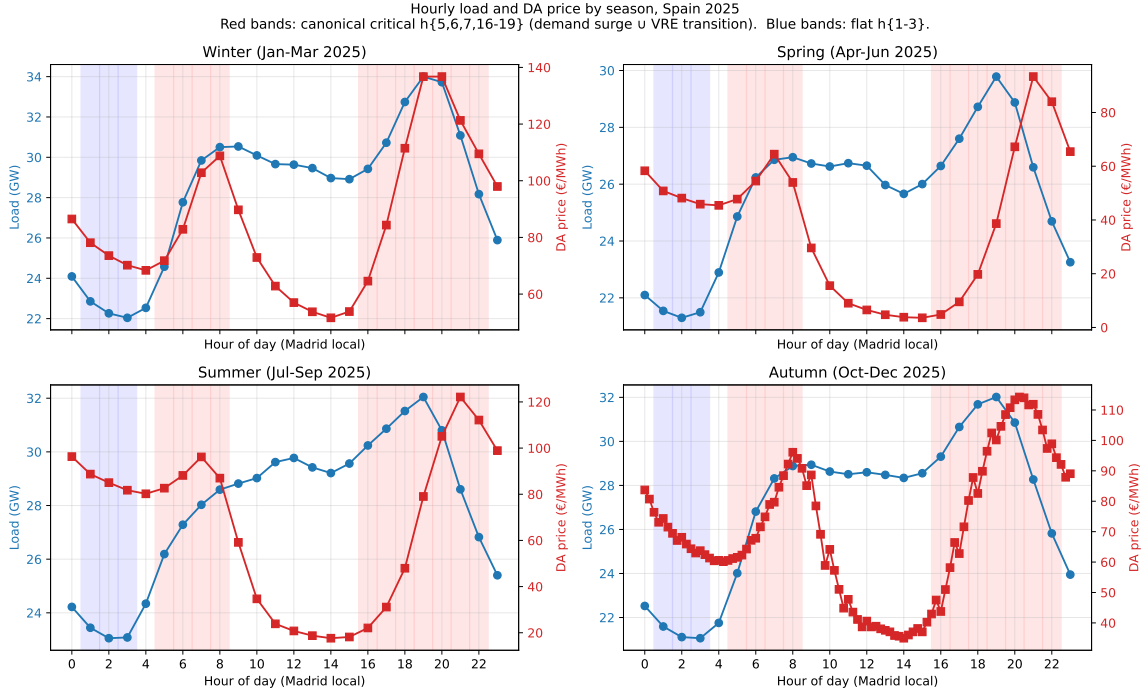


Figure 1: Hourly load and DA price by season, Spain 2025. Blue: demand; red: DA clearing price. Red bands: critical hours (05:00–09:00, 16:00–23:00); blue bands: flat hours (01:00–04:00).

Reading (supportive). Critical hours combine the morning ramp and evening peak; flat hours fall in the overnight valley. The pattern is stable across seasons.

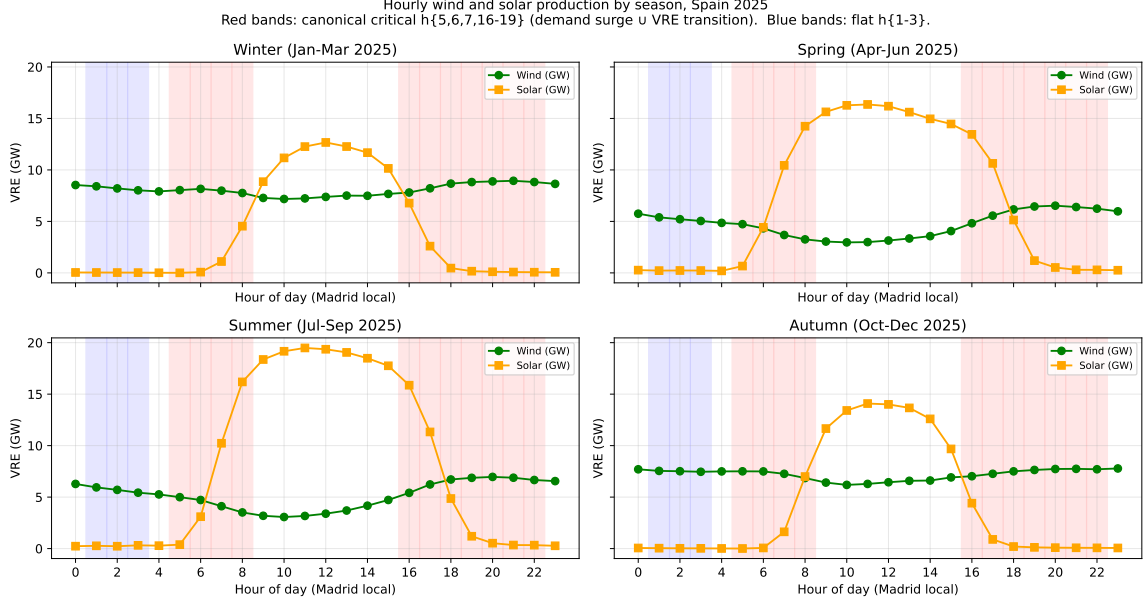


Figure 2: Hourly wind and solar production by season, Spain 2025. Red bands: critical hours; blue bands: flat hours.

Reading (supportive). Solar production concentrates around midday and fades sharply at the critical-hour boundaries; wind is broadly flat. Critical hours systematically face low renewable supply, reinforcing their role as the time-varying window.

3.5 Firms in the analysis sample

MENTION THAT WE USE LISTA_UNIDADES AND LISTA_AGENTES FROM OMIE, AS WELL AS THE SPECIFIC FAMILIES FROM OMIE, WHICH HAS FIRM AND UNIT INDICATORS. ALSO IMPORTANT TO MENTION THE CNMC CLASSIFICATION OF DOMINANT (SOURCE)

The empirical analysis follows nine parent firms in two blocks. The first block contains the *dominant operators in Spanish generation* (the CNMC *operador-dominante-en-generación* tier): Iberdrola, Endesa, Naturgy and EDP-Spain. EDP-Portugal joins this block as the dominant Iberian generator on the Portuguese side of MIBEL. The second block contains four other large vertically-integrated operators – Repsol, Engie España, TotalEnergies, and Moeve – which retain substantial portfolios but are non-dominant in the CNMC sense. Whether the firms in each block actually have a strategic margin in critical hours is an empirical question that we answer in Section 4.1, examining pivotality on the day-ahead bid data.

Table 1: Firms in the analysis sample. Capacity: max DA-offered power across 2025, share-weighted. Market share: PDBF programmed sell, 2025. Channel breakdown in Appendix B.2.

Parent firm	Main generation technologies	Capacity (GW)	Market share (%)
<i>Dominant operators in Spanish generation (CNMC tier)</i>			
Iberdrola	Solar, Hydro, CCGT	31.03	24.94
Endesa	Hydro, CCGT, Nuclear	12.46	16.39
Naturgy	CCGT, Hydro, Solar	12.60	12.34
EDP-Spain	Wind, Hydro, Solar	4.73	4.14
EDP-Portugal	Hydro, CCGT	7.42	15.17
<i>Other large operators (non-dominant)</i>			
Repsol	Solar, Hydro	0.61	1.43
Engie España	Wind, Solar, CCGT	4.05	1.73
TotalEnergies	Solar, CCGT, Wind	1.19	0.63
Moeve	Solar, Hydro, CCGT	1.28	0.01

Reading (descriptive). The four CNMC-dominant operators (Iberdrola, Endesa, Naturgy, EDP-Spain) plus EDP-Portugal together hold the bulk of Iberian generation capacity. The non-dominant block (Repsol, Engie, TotalEnergies, Moeve) is substantially smaller and serves as the placebo.

3.6 Identification strategy and treatment-effect set

WE SHOULD DEFINE HERE HOW WE DEFINE PIVOTALITY AND HOW WE CALCULATE IT, WITH DISCUSSION.

ALSO WE SHOULD DISCUSS THAT THE TREATMENT IN OUR IDENTIFICATION IS AT THE HOUR LEVEL (CRITICAL VS FLAT), NOT AT THE FIRM LEVEL, AND THAT FIRMS ENTER THE ANALYSIS AS SUBJECTS: PIVOTAL FIRMS ARE THOSE EXPECTED TO REACT TO THE TREATMENT BECAUSE THEY HAVE A STRATEGIC MARGIN IN CRITICAL HOURS; NON-PIVOTAL FIRMS SERVE AS PLACEBO BECAUSE THEY BID AT MARGINAL COST REGARDLESS OF HOUR.

HOWEVER, WE SHOULD THINK IF THIS IS THE BEST WAY TO PRESENT IT, MAYBE IT IS BETTER TO TALK ABOUT "EXPOSURE" TO THE TREATMENT ACROSS TIME OR SOMETHING LIKE THAT!! TO THINK ABOUT IT

3.7 Parallel-trends discussion

IMPORTANT THING HERE IS TO DISCUSS IT FOR EACH OF THE OUTCOMES AND TO TALK ABOUT CONDITIONAL PARALLEL TREND WITH RENEWABLE CAPACITY CONTROLS!

GAS PRICES CAN AFFECT DIFFERENTLY CRITICAL AND FLAT HOURS, BUT WE SEE THAT GAS IS PIVOTAL IN THE SAME WAY FOR BOTH HOUR CLASSES, SO IT IS NOT A CONCERN FOR IDENTIFICATION.

Table 2: Price-setting technology by hour-class, October–December 2025 (post-MTU15-DA). For each (date, period) the price-setting technology is the one whose accepted sell tranche has the highest bid price (i.e., the tranche that determines the uniform clearing price). Table reports the share of (date, period) cells in each hour-class where each technology is the price-setter.

Technology	Critical	Midday	Flat	Crit–Flat
Hydro	58.7	24.3	57.8	+0.8
Hydro_pump	19.0	1.9	13.7	+5.4
CCGT	9.1	3.0	9.0	+0.2
Wind	4.5	21.1	9.7	-5.2
Solar PV	2.2	16.1	0.0	+2.2
Cogen	1.7	11.4	3.0	-1.3
Hydro_RES	1.7	2.6	2.2	-0.6
Biomass	0.8	4.4	1.8	-1.0
Hybrid_RES	0.8	7.5	0.8	-0.0
Import	0.3	0.0	1.4	-1.1
RE_Tarifa_CUR	0.3	6.6	0.1	+0.2

Table 3: Price-setting technology by hour-class, October–December 2024 (pre-MTU15-DA). Same construction as Table 2.

Technology	Critical	Midday	Flat	Crit–Flat
Hydro	62.2	35.6	55.5	+6.7
CCGT	13.8	17.1	18.4	-4.6
Hydro_pump	11.8	1.0	5.9	+5.9
Wind	4.0	16.2	7.7	-3.7
Hydro_RES	2.2	2.9	2.7	-0.5
Cogen	1.6	4.0	6.2	-4.6
Solar PV	1.5	12.1	0.7	+0.9
Coal	1.0	0.4	0.0	+1.0
Import	0.6	0.0	1.1	-0.5
Hybrid_RES	0.5	6.9	0.4	+0.1
Biomass	0.4	1.4	1.5	-1.1
RE_Tarifa_CUR	0.3	1.8	0.0	+0.3

Reading (supportive). Hydro is the dominant price-setter in both hour-classes ($\sim 58\%$ of periods post-reform), reflecting reservoir water-value sitting at the merit-order edge. CCGT sets the price in 9.1% of critical-hour periods vs 9.0% of flat-hour periods post-reform — a 0.2 pp critical–flat gap, essentially symmetric. A TTF (gas-price) shock therefore propagates into both hour-classes through the same channel and at the same intensity; an asymmetric gas confound on the critical–flat clearing-price spread would have required CCGT to set the price disproportionately in one class, which we do not observe. Hydro_pump’s critical–flat gap is +5.4 pp (19.0% vs 13.7%) post-reform; this is consistent with regular storage arbitrage (charge during low-price midday/flat hours, discharge into the evening peak) and is not, on its own, evidence of MTU15-induced strategic behaviour. (Coverage caveat: $\sim 36\%$ of critical-hour periods have their clearing price set by a unit not present in our generation-unit panel `lista_unidades.csv` — e.g. Portuguese-zone generation units, demand-side buy bids that set the clearing, or block-order/interconnection bids without a single unit code — and are excluded from the table. *Bilateral contracts do not enter EUPHEMIA*: per the OMIE file specification (`ficherosomie137.pdf`, §5.1.2.1, §5.1.2.3), bilateral-contract executions are reported separately by the unit owners *after*

clearing and merged into the final daily program (PDBF, `offer_type=4`), so they cannot set the auction price. The patterns above describe the remaining matched majority of periods.)

4 Bidding-behavior evidence of the mechanism

4.1 Pivotality: which firms can exploit the granularity reform

OJO, HYDRO DOES NOT EXPLOIT GRANULARITY BECAUSE THE DISPATCH DECISION IS DOMINATED BY RESERVOIR WATER-VALUE OPTIMIZATION (A MULTI-DAY PROBLEM), NOT WITHIN-HOUR STRATEGY. SO THE GRANULARITY REFORM'S "STRATEGIC EXPLOITATION CHANNEL" HAS NO PURCHASE HERE EVEN THOUGH THE CAPACITY FOR MARKET POWER EXISTS!!!!

ONLY CCGT EXPLOITS GRANULARITY!! AND IN PARTICULAR NATURGY!

Table 4: Pivotality by parent firm and hour-class, CCGT, October–December 2025. Pivotality = share of (date, unit, period) cells with a bid tranche within 1 EUR/MWh of DA clearing.

Firm (parent)	flat (01:00–04:00)	critical (05:00–09:00, 16:00–23:00)	Δ crit–flat
<i>Tightly pivotal in critical hours (within 1 EUR/MWh of clearing)</i>			
Naturgy	0.006	0.044	0.038
EDP-Spain	0.026	0.041	0.015
EDP-Portugal	0.005	0.028	0.023
Iberdrola	0.003	0.012	0.009
Endesa	0.001	0.004	0.003
<i>Essentially never tightly pivotal (negative control)</i>			
Engie España	0.000	0.000	0.000
Moeve (formerly Cepsa)	0.000	0.000	0.000
Repsol	0.000	0.000	0.000
TotalEnergies	0.000	0.000	0.000

Reading (supportive). Naturgy is the most pivotal firm in critical hours (4.4%), followed by EDP-Spain and EDP-Portugal. Repsol, Engie, TotalEnergies, and Moeve are essentially never tightly pivotal and serve as the negative control.

4.2 Per-firm bid-shape and granularity exploitation

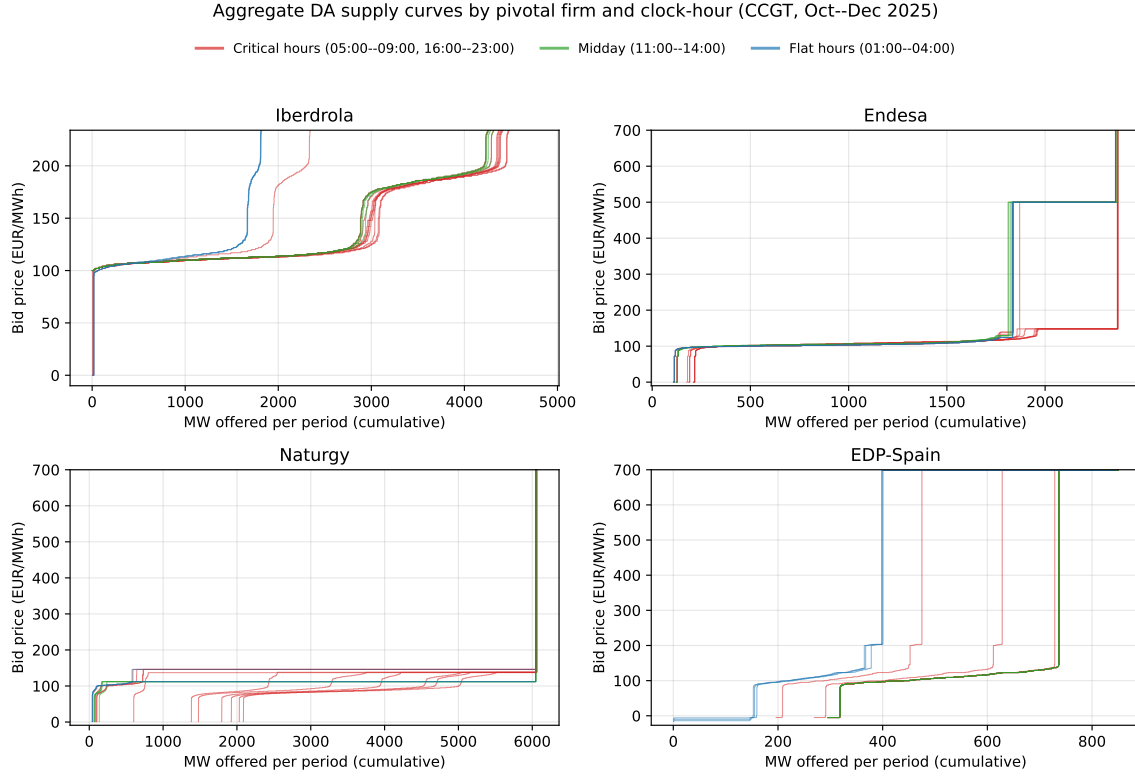


Figure 3: Aggregate DA supply curves by pivotal firm and clock-hour, CCGT, October–December 2025. EUPHEMIA aggregation (public description, sec. 7.1) per (firm, hour): pool sell tranches, sort ascending by price, cumulate, normalise by $N_{\text{dates}} \times 4$ quarters. Red: 11 critical hours; green: 4 midday hours (11:00–14:00, dropped from the DiD); blue: 3 flat hours. Submission rates by hour-class in Table 7; Hydro analogue in Appendix A.11.

Reading (supportive). Within-day variation is what we look at: the spread of red curves within a panel shows whether the firm differentiates its bid hour-by-hour in critical hours. Iberdrola’s red curves spread across a wide range while the three blue flat-hour curves sit tightly together; midday (green) sits between the two. Naturgy’s pattern is sharper: critical-hour curves span $\sim 1,500$ – $6,000$ MW at moderate prices, flat curves cluster at the lower edge. EDP-Spain shows the same within-class fan in critical hours. Endesa is the exception: red, green, and blue curves overlap and do not spread — no within-day differentiation.

Table 5: Per-firm DA bid-shape detail, October–December 2025. Mean number of tranches, median tranche price, total quantity offered, and the share of clock-hours in which all four 15-min quarters carry an identical bid (mechanical-repeat rate).

Firm	Hour-class	Mean tranches	Median price (EUR/MWh)	Total qty (MW)	Mechanical repeat rate	N
Iberdrola	Critical	9.70	134.8	612.4	0.990	31,324
	Flat	4.38	245.7	614.4	0.969	8,556
Endesa	Critical	1.92	1090.0	460.7	0.982	20,780
	Flat	1.82	1179.4	460.6	1.000	5,652
Naturgy	Critical	6.30	538.1	410.4	0.811	59,720
	Flat	1.32	945.5	410.4	1.000	16,260
EDP-Spain	Critical	5.76	121.1	426.8	1.000	8,052
	Flat	3.68	375.9	424.9	1.000	2,196
EDP-Portugal	Critical	6.60	153.0	407.0	0.075	18,556
	Flat	5.52	233.0	407.0	0.270	5,064

Reading (supportive). Ladder enrichment in critical hours is clear in Iberdrola (9.7 vs 4.4 tranches), Naturgy (6.3 vs 1.3) and EDP-Spain (5.8 vs 3.7). The mechanical-repeat rate drops in critical hours for Naturgy (0.81 vs 1.00) and EDP-Portugal (0.07 vs 0.27): they are the firms varying their bids across the four quarters of an hour. The higher flat-hour median prices are mechanical — low overnight demand pushes units into precautionary, near-cap bids regardless of strategy.

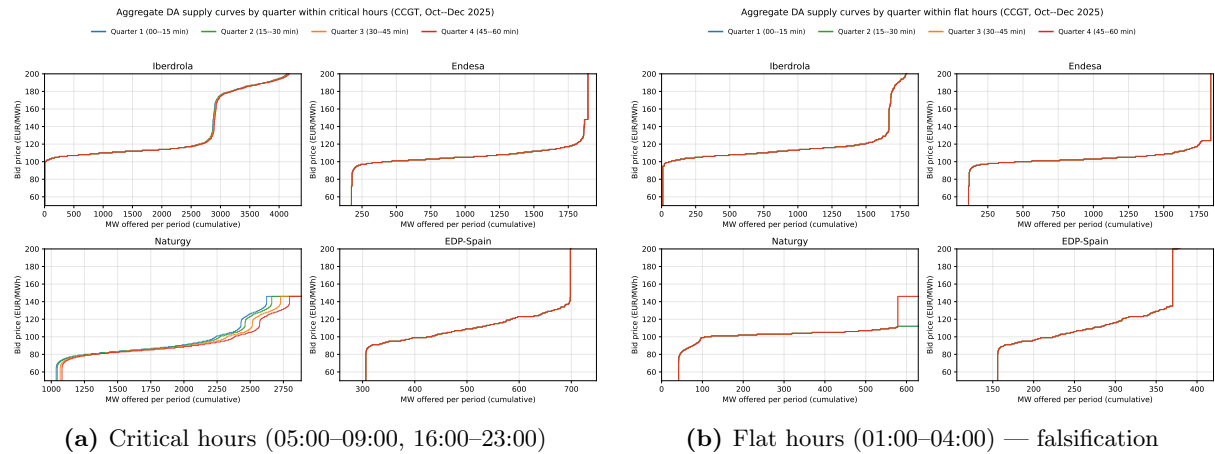


Figure 4: Aggregate DA supply curves by quarter, CCGT. EUPHEMIA aggregation per (firm, quarter) over October–December 2025. Y-axis zoomed to 50–200 EUR/MWh, the clearing-relevant band. One curve per quarter. The left panel restricts to the 11 critical hours; the right panel to the 3 flat hours and serves as a falsification check.

Reading (supportive). In critical hours Naturgy shows visible quarter dispersion in the 1,500–2,800 MW band (Q1 below Q4 in the body of the curve); in flat hours the four quarters of Naturgy — and of every other firm — collapse onto a single curve. Granularity differentiation is hour-class-specific: it appears only where within-hour demand variation gives it economic content. This is the falsification check the within-day design requires.

IDA mirror. The same construction applied to the intraday auction (ICAB + IDET, pooled across the 3 daily IDA sessions) is shown in Figures 5–6b.

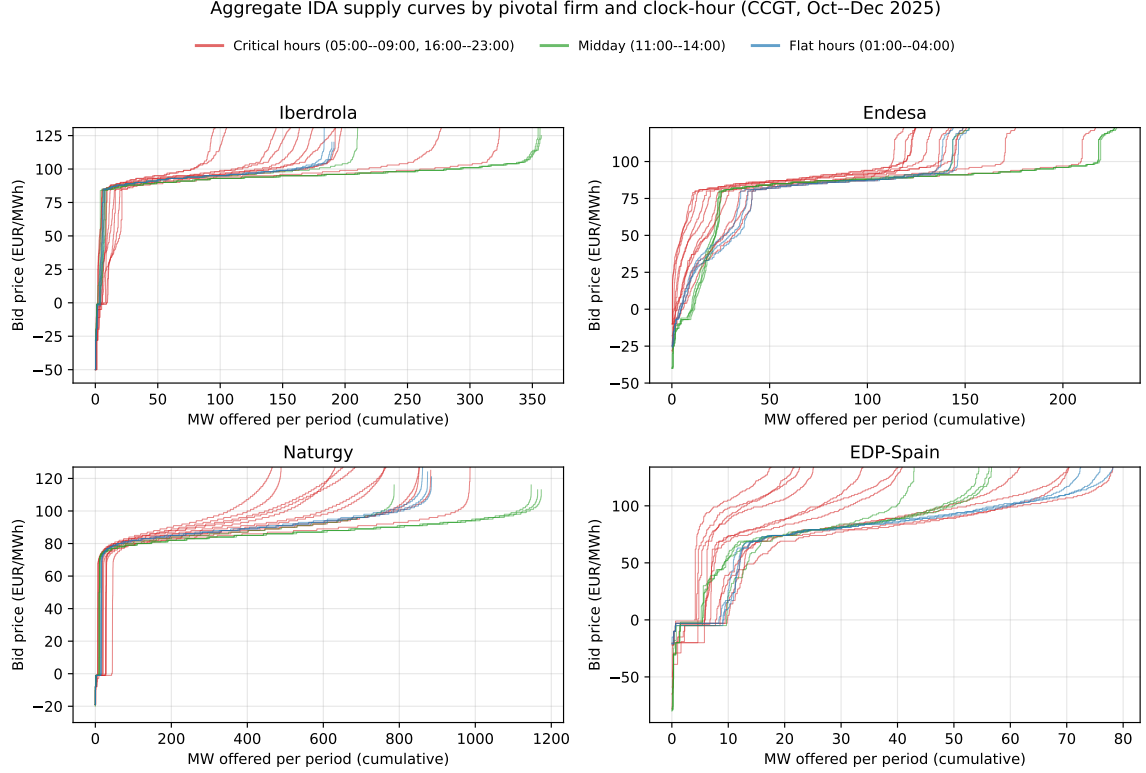


Figure 5: Aggregate IDA supply curves by pivotal firm and clock-hour, CCGT. Same construction as Figure 3 on ICAB+IDET data, pooled across the 3 IDA sessions per day; normalisation $N_{\text{dates}} \times 4 \times 3$.

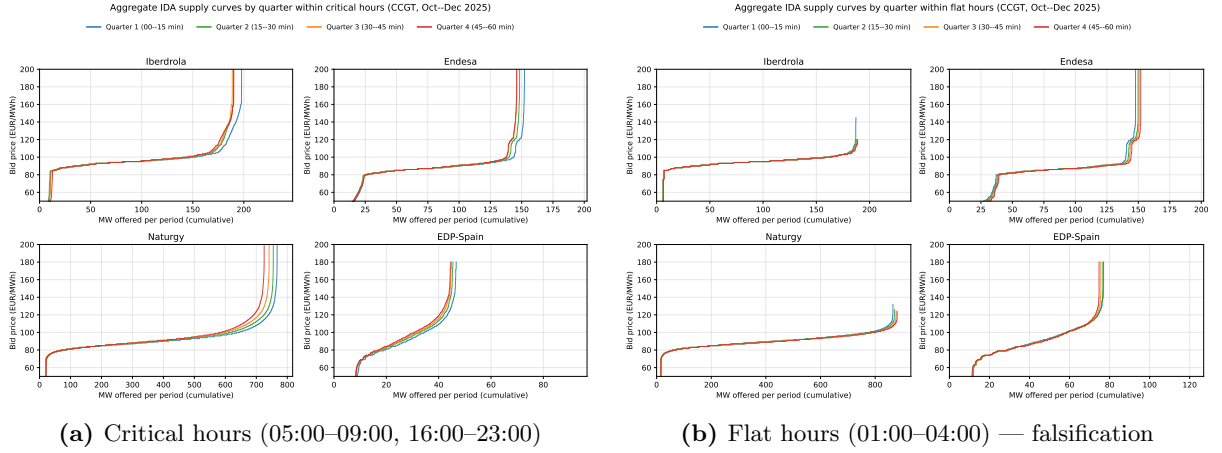


Figure 6: Aggregate IDA supply curves by quarter, CCGT. Same construction as Figure 4 on ICAB+IDET data, pooled across the 3 IDA sessions per day. Left panel: critical hours; right panel: flat hours (falsification).

Reading (supportive). In IDA, all four pivotal firms (not just Naturgy as in DA) show monotone Q1–Q4 dispersion in critical hours: Q1 offers the most MW, Q4 the least, at prices near clearing. The flat-hour falsification holds for every firm. IDA is where pivotal firms actively reposition q_2 , and the granularity exploitation pattern is sharper here than in the day-ahead.

4.3 Demand-side and storage bidding: who exploits granularity

Mirror of the per-firm supply-side analysis on the buy side and on pumped-storage discharge. EUPHEMIA stepwise aggregation (OMIE convention; [docs/omie/euphemia_functioning_1812.pdf](#), slide 7): demand orders are sorted price-descending and cumulated, so the resulting curve slopes downward in (Q, P) -space. We focus on pump-storage charge bids (tech-group Pump_load) as the headline because pump-storage is the textbook arbitrage tool — the technology where the strategic-granularity story predicts the sharpest within-day price discrimination.

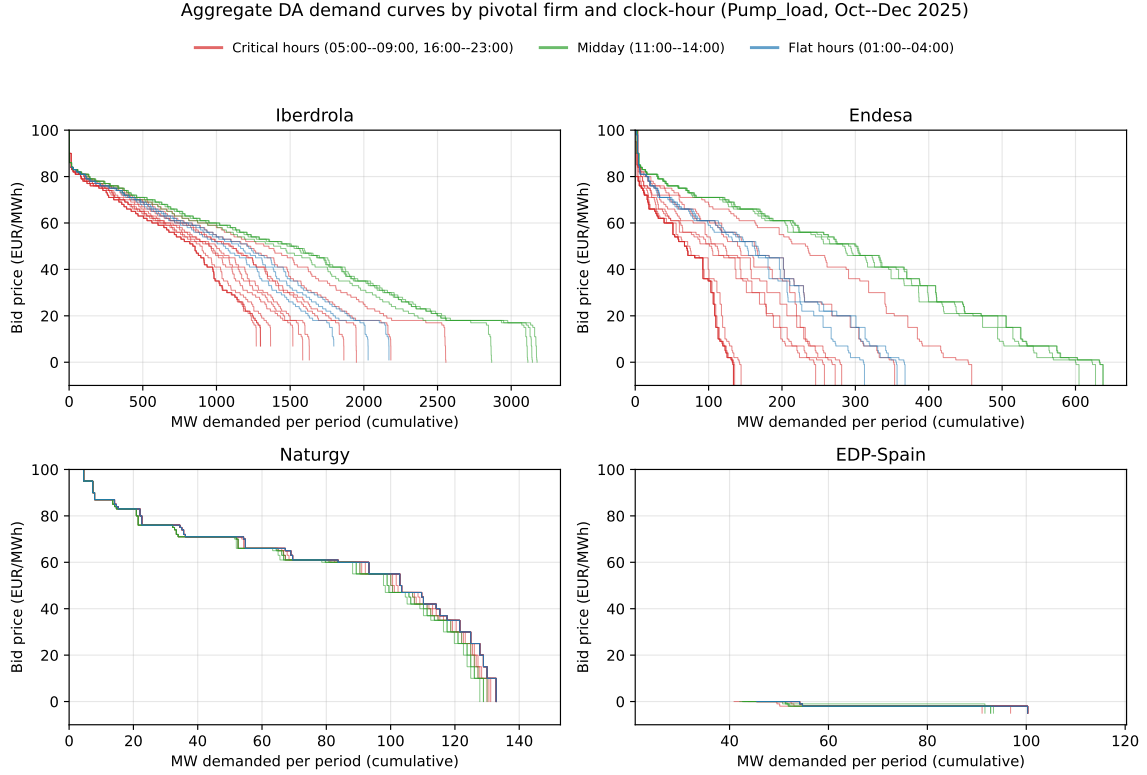


Figure 7: Aggregate DA demand curves by pivotal firm and clock-hour, pump-storage charge bids, October–December 2025. Each step is the EUPHEMIA aggregation of buy tranches at each integer-rounded limit price, cumulated price-descending and normalised by $N_{\text{dates}} \times 4$ periods per clock-hour. Critical hours red, midday green, flat hours blue.

Reading (mixed). Iberdrola and Endesa show clear hour-of-day discrimination: midday (green) curves shifted right of flat (blue) curves, both shifted right of critical (red). Both firms charge more pump-storage volume at higher willingness-to-pay during midday, consistent with absorbing cheap solar output, and shrink their charge volumes in critical hours when the marginal value of selling exceeds the storage round-trip return. Naturgy shows no hour-class discrimination — the three lines overlap. EDP-Spain has only a small pump-storage book and bids near zero EUR/MWh.

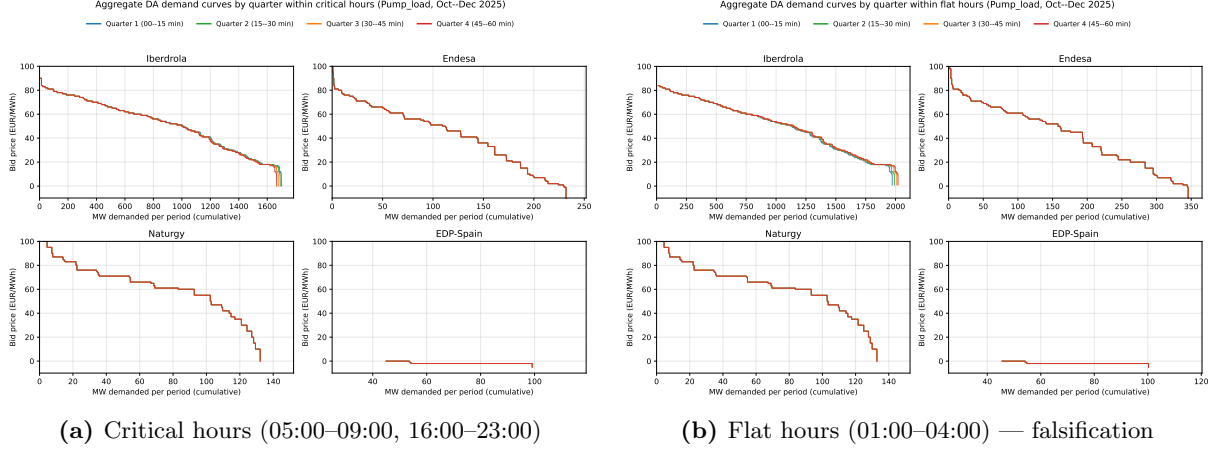


Figure 8: Aggregate DA demand curves by quarter, pump-storage charge bids, October–December 2025. Same construction as Figure 7; one curve per 15-min quarter. Left panel: critical hours; right panel: flat hours (falsification).

Reading (supportive of null). The four quarter curves overlap almost exactly for every firm in both critical and flat hours: pump-storage operators do not differentiate their charge bids across the 15-min quarters of a clock-hour. The within-hour granularity available since MTU15-DA is unused on this margin — even Iberdrola and Endesa, who discriminate strongly at the clock-hour level, replicate identical bids across the four quarters. This pins the within-hour granularity-exploitation effect to the supply side (pivotal dispatchable units), not to demand-side or storage charging strategy.

IDA mirror. The same construction applied to the intraday auction (ICAB + IDET, pooled across the 3 daily IDA sessions) is shown in Figures 9–10.

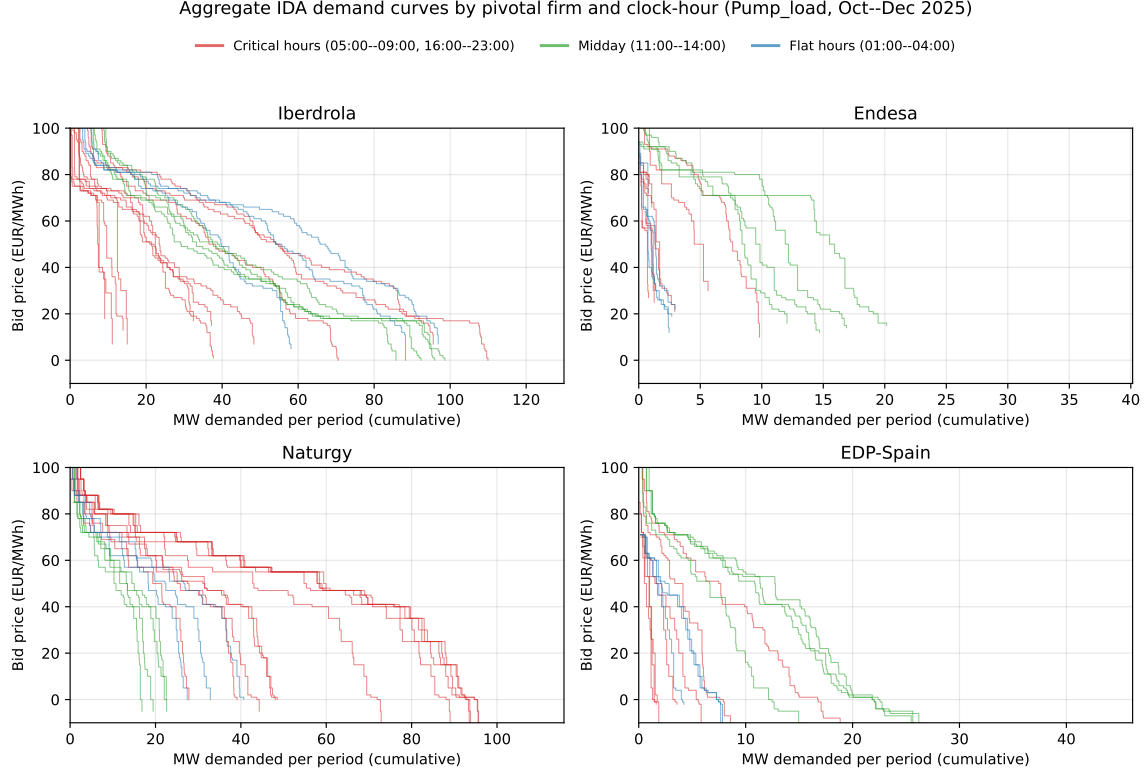


Figure 9: Aggregate IDA demand curves by pivotal firm and clock-hour, pump-storage charge bids, October–December 2025. Same construction as Figure 7 on ICAB+IDET data, pooled across the 3 IDA sessions per day; normalisation $N_{\text{dates}} \times 4 \times 3$.

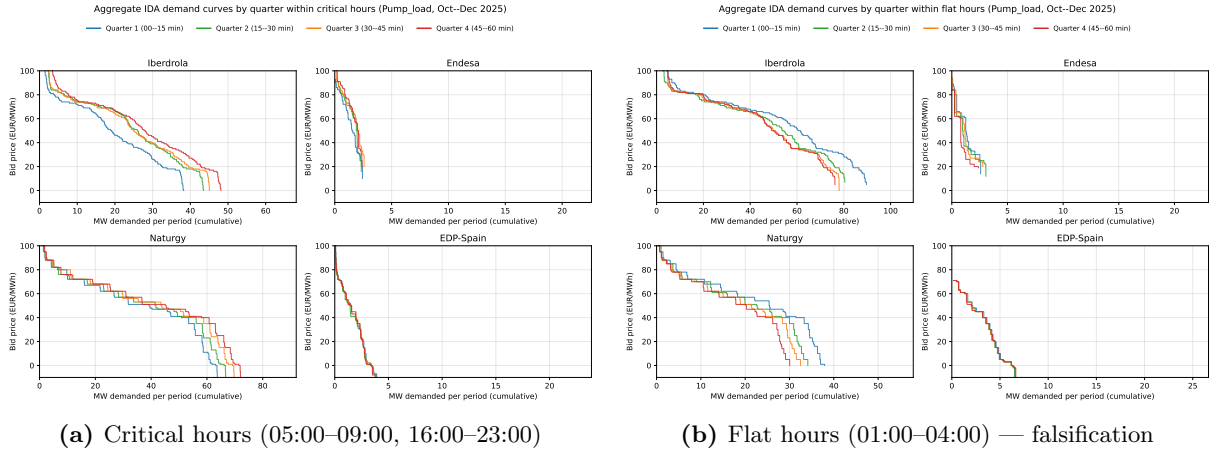


Figure 10: Aggregate IDA demand curves by quarter, pump-storage charge bids, October–December 2025. Same construction as Figure 8 on ICAB+IDET data, pooled across the 3 IDA sessions per day. Left panel: critical hours; right panel: flat hours (falsification).

Compact grids covering all demand-side technologies (Pump_load, Retailer, Direct_consumer) are in Appendix A.12 (DA) and Appendix A.14 (IDA); the storage-discharge mirror (Hydro_pump, Hybrid_RES_storage) is in Appendix A.13 (DA) and Appendix A.15 (IDA).

4.4 Operational vs strategic use of MTU15 granularity

TWO CHANNELS BY WHICH A FIRM CAN “USE” WITHIN-HOUR GRANULARITY (SEE §1):

- OPERATIONAL: TRACK PHYSICAL VARIATION OF OWN PRODUCTION ACROSS QUARTERS (WIND / SOLAR FORECAST UPDATES, RAMPING). BID QUANTITY VARIES; BID PRICE STAYS AT MARGINAL COST.
- STRATEGIC: RE-PRICE ACROSS QUARTERS TO EXTRACT RENTS FROM WITHIN-HOUR DEMAND SWINGS. BID PRICE VARIES IN A DEMAND-CORRELATED WAY.

THE WITHIN-HOUR QUARTER-DISSIMILARITY TEST IN APPENDIX A.16 (FRACTION OF (FIRM, UNIT, DATE, HOUR) CELLS WHERE THE FOUR 15-MIN QUARTERS' BIDS ARE NOT BIT-IDENTICAL) DOES NOT SEPARATE THESE TWO CHANNELS. THE CRITICAL-FLAT ASYMMETRY DOES: STRATEGIC RE-PRICING REQUIRES SYSTEMATIC WITHIN-HOUR DEMAND VARIATION (CRITICAL HOURS BY OUR DEFINITION) AND SHOULD BE ABSENT WHEN SUCH VARIATION IS LOW (FLAT HOURS). OPERATIONAL REFINEMENT IS DRIVEN BY OWN-PRODUCTION VARIABILITY, WHICH IS NOT TIED TO HOUR-CLASS.

Table 6: Critical-flat asymmetry of the within-hour bid-curve dissimilarity, October–December 2025. Each row stratifies the dissimilarity statistic $D = \max_{i < j} W_1(\mu_i, \mu_j)$ over the four 15-min quarters of a clock-hour: D is the L1 area between cumulative bid curves of any pair of quarters — a 1-D Wasserstein-1 (Earth Mover’s Distance, Vallender, 1973) on the bid measure, defined in Equation 4. Entries report the percentage of (firm, unit, date, hour) cells with $D > 0$ in critical vs flat hours, and the gap. Strategic re-pricing should appear as a positive critical-flat gap; operational refinement (e.g. wind/solar forecast updates) as a near-zero gap.

Channel	Tech	Firm	Critical	Flat	Crit – Flat
Strategic	CCGT	GN	24.5	0.0	+24.5
Strategic	CCGT	IB	5.8	3.9	+1.9
Operational	Wind	IB	97.2	97.3	−0.1
Operational	Wind	GN	92.7	92.6	+0.1
Operational	Wind	HC	93.7	93.8	−0.1
Operational	Solar PV	GN	15.9	0.0	+15.9
Operational	Retailer	GE	76.5	76.3	+0.2
Operational	Retailer	IB	40.9	40.2	+0.7
Operational	Direct_consumer	IB	96.0	95.3	+0.7
Mechanical	Pump_load	IB	3.2	5.5	−2.3
Mechanical	Pump_load	GE	0.0	0.0	0.0
Mechanical	Hydro_pump	IB	0.4	0.0	+0.4
Mechanical	Hydro_pump	GE	0.0	0.0	0.0

Reading (supportive). Only Naturgy CCGT shows the strategic signature (large positive crit-flat gap). Renewables (Wind, Solar PV) and pivotal-firm retail demand vary across quarters in essentially every cell, but symmetrically across hour-classes — consistent with operational forecast / consumption refinement, not market-power exploitation. Pump-storage is genuinely mechanical on both margins. The within-hour strategic-exploitation channel of MTU15 is therefore narrow: pivotal dispatchable thermal in hours with material within-hour demand variation.

5 Main empirical results

5.1 Strategic withholding and intraday upward repositioning

MECHANISM: UNDER THE HIGHER-GRANULARITY REGIME, A PIVOTAL DOMINANT FIRM IN A CRITICAL HOUR CAN WITHHOLD CAPACITY FROM THE DAY-AHEAD CLEARING AND REPOSITION THAT CAPACITY UPWARD IN THE INTRADAY AUCTIONS AT HIGHER PER-QUARTER PRICES. UPWARD SELL ADJUSTMENT SUMMED ACROSS THE THREE INTRADAY AUCTIONS, $q_{2,fdh}$, IN MWH PER UNIT-CLOCK-HOUR:

$$q_{2,fdh} = \alpha + \beta_1 \text{crit}_h + \beta_2 \text{post}_d + \beta_3 (\text{crit}_h \times \text{post}_d) + \gamma_f + \delta_{\text{DOW}(d)} + \varepsilon_{fdh}, \quad (2)$$

with firm fixed effects γ_f , day-of-week fixed effects $\delta_{\text{DOW}(d)}$, and cluster-robust standard errors by date. The sample restricts to the same-calendar-month windows October–December 2024 (pre) and October–December 2025 (post), so the March 2025 intraday reform is held constant. Critical hours are 05:00–09:00 and 16:00–23:00; flat hours are 01:00–04:00. Pivotal firms are those with non-trivial price-setting capacity in critical hours (Iberdrola, Endesa, Naturgy, EDP-Spain, EDP-Portugal); non-pivotal firms (Repsol, Engie España, TotalEnergies, Moeve) serve as a negative control.

5.2 Parallel trends discussion

$$\mathbb{E}[Y_{\text{post}}(0) - Y_{\text{pre}}(0) \mid \text{crit}] = \mathbb{E}[Y_{\text{post}}(0) - Y_{\text{pre}}(0) \mid \text{flat}]. \quad (3)$$

6 Conclusion

A Robustness checks

A.1 Sensitivity to critical-hours definition

A.2 Sensitivity to firm-partition (pivotality vs administrative)

A.3 Same-cal-month vs full-window comparisons

A.4 β_3 stratified by technology

A.5 Per-firm decomposition: q_2 and day-ahead cleared MWh

A.6 DA $\rightarrow$ IDA price wedge (supporting outcome)

Operación reforzada as a non-collinear regulatory overlay. REE has held a continuous *operación reforzada* stance since the 28 April 2025 Iberian blackout: a real-time technical-restrictions regime (PO-3.x cascade, PO-7.4 zonal reactive market from June 2025) that activates additional reserves at the unit level and can lift the day-ahead-to-intraday wedge. Figure 11 shows the time profile.

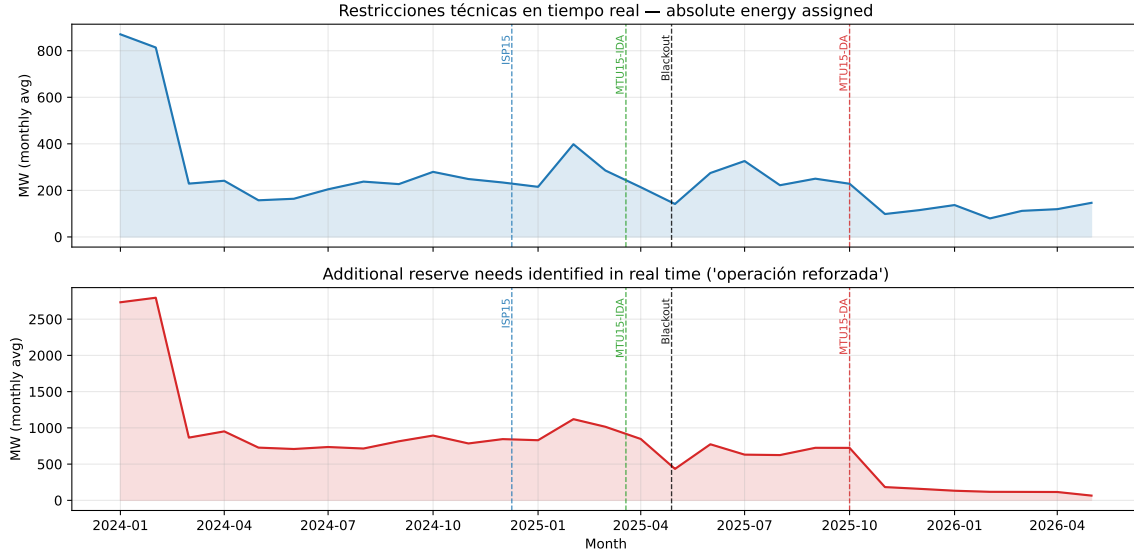


Figure 11: Reforzada / RT2 intensity, January 2024 – May 2026. Top panel: monthly average of absolute energy assigned via real-time technical restrictions (ESIOS indicator 10052). Bottom panel: monthly average of additional reserve needs identified in real time (ESIOS indicator 1880; coverage from 2021-05). Dashed vertical lines mark, in order, ISP15 (2024-12-09), MTU15-IDA (2025-03-19), the Iberian blackout (2025-04-28) and MTU15-DA (2025-10-01).

Reading. RT2 activation steps up sharply at the blackout and then *remains continuously high* through both the DA60/ID15 regime (April–September 2025) and the post-MTU15-DA window (October 2025 onward). It is therefore a regime change that begins five months before MTU15-DA and persists across it: $D_{\text{MTU15-DA}}$ and $D_{\text{reforzada}}$ are not collinear in panels including the 2025-03-19 to 2025-04-27 sub-window (post-MTU15-IDA, pre-reforzada). Within-day DiD with calendar-month FE in the post-reforzada window absorbs the level of reforzada activation; differences in MTU15-DA exposure across critical-vs-flat hours within a calendar month identify the reform effect cleanly.

A.7 Operational vs strategic decomposition (PIBCA vs PHF)

A.8 Bid-shape robustness in the competitive-zone restriction

A.9 DA submission rates by hour-class (selection check)

Table 7: Mean per-unit CCGT DA submission rate, by firm and hour-class, October–December 2025.

Firm	Critical	Midday	Flat	N units
Iberdrola	0.860	0.861	0.861	9
Endesa	0.733	0.734	0.731	7
Naturgy	0.868	0.868	0.866	17
EDP-Spain	0.995	0.995	0.995	2
EDP-Portugal	0.917	0.915	0.917	5

Reading (supportive). Within each firm, critical / midday / flat rates differ by < 0.003 . No hour-class selection: same units bid across hour classes.

A.10 IDA per-firm supply curves, all technologies

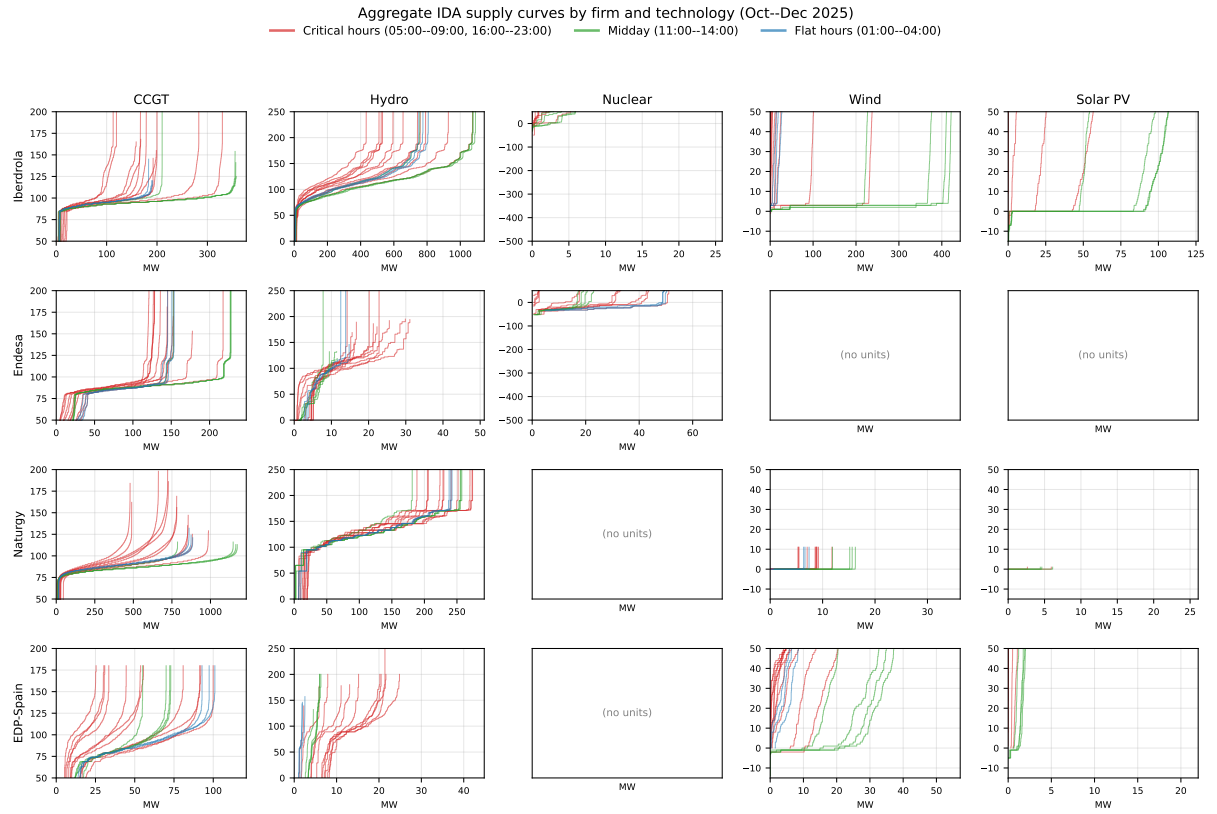


Figure 12: Aggregate IDA supply curves by firm and technology, by hour-class. Compact grid analogue of Figure 14 on IDA data.

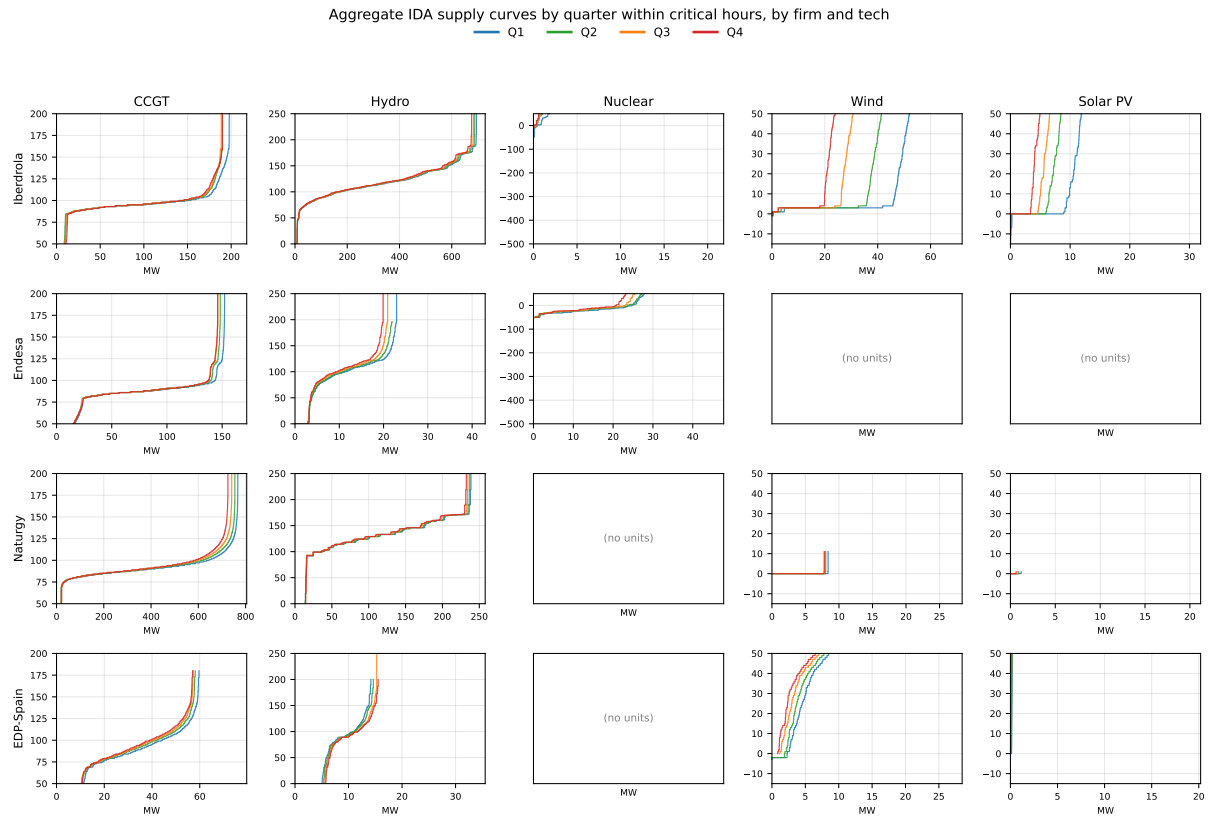


Figure 13: Aggregate IDA supply curves by quarter within critical hours, by firm and technology.

Reading (supportive). The CCGT column carries the quarter dispersion across all four firms (visible in the Q1–Q4 colour fan). Hydro shows the same effect at smaller scale. Wind, Solar PV and Nuclear columns are flat across quarters as expected.

A.11 Per-firm aggregate supply curves, all technologies (DA)

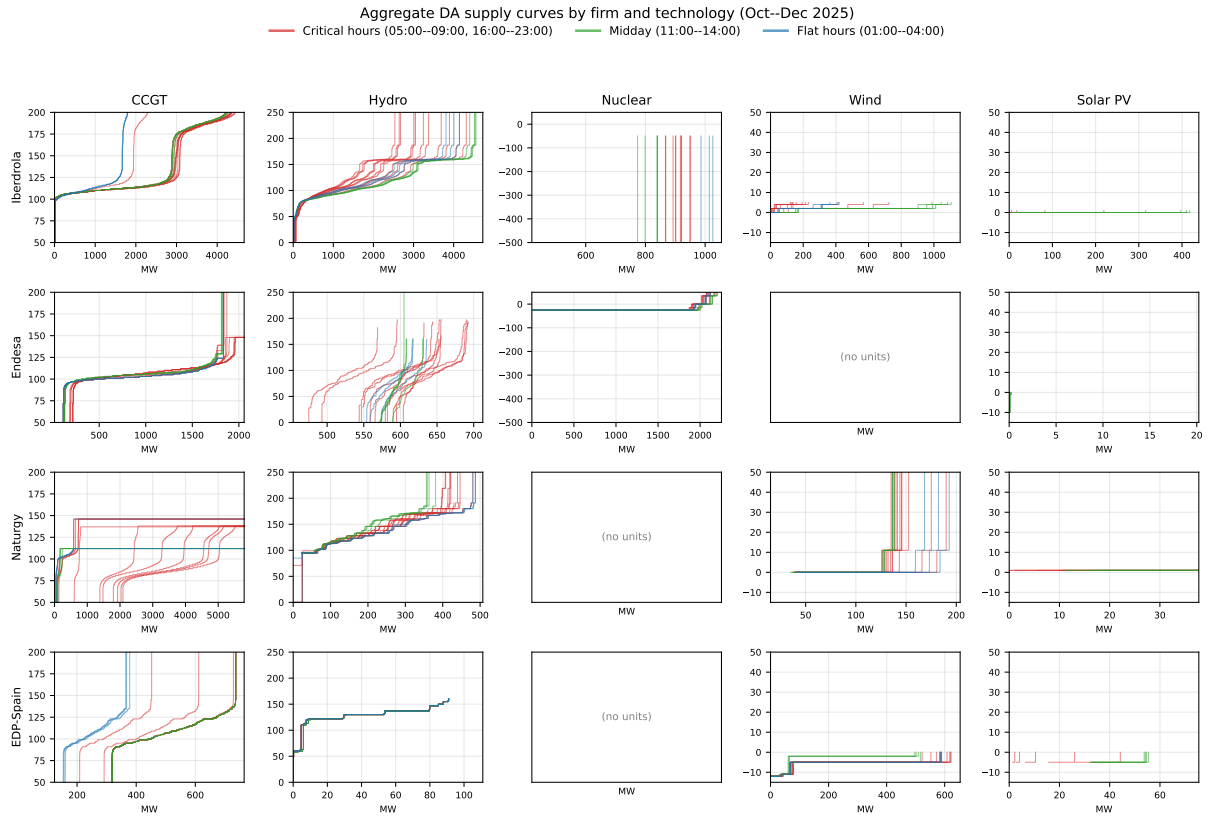


Figure 14: Aggregate DA supply curves by firm and technology, October–December 2025, by hour-class. Same construction as Figure 3. Adaptive y-axis per panel: focus on the 80th percentile of quantity-weighted price.

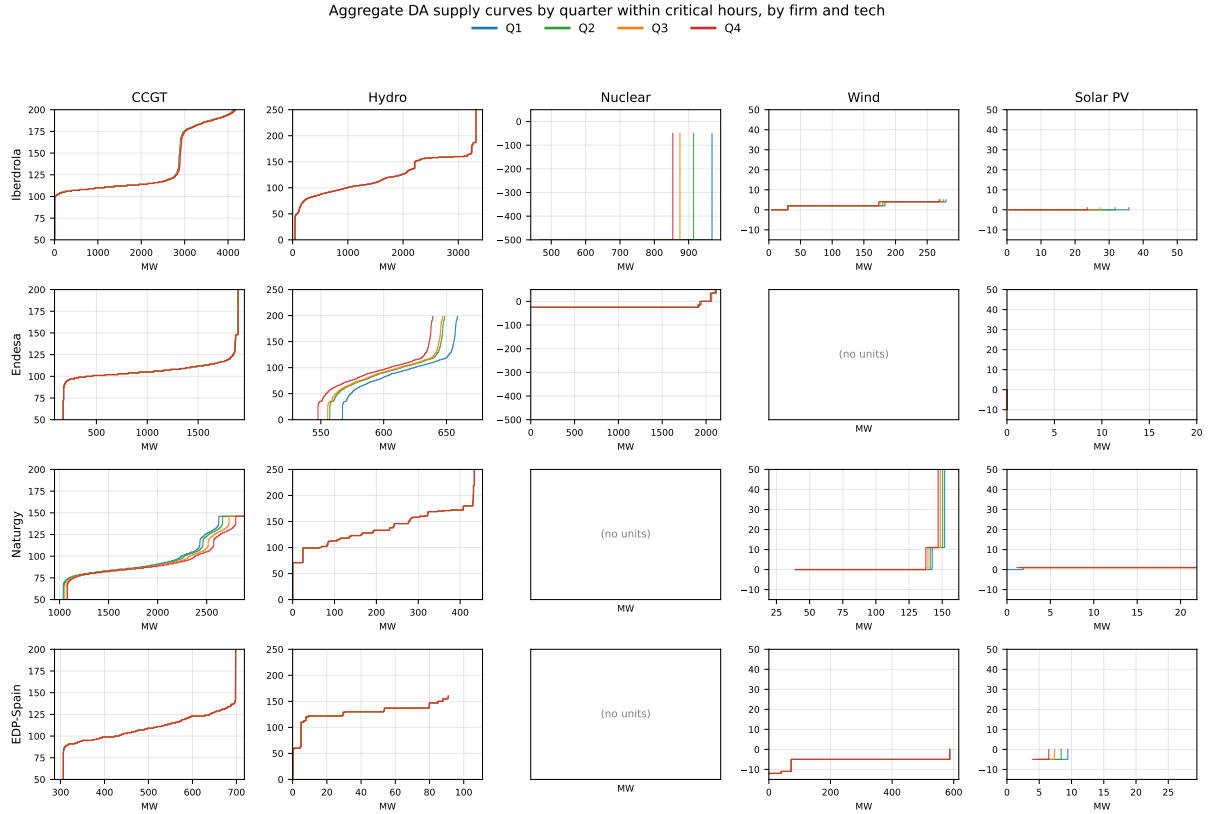


Figure 15: Aggregate DA supply curves by firm and technology, October–December 2025, by quarter within critical hours. Same construction as Figure 4a. Y-axis clamped to 50–200 EUR/MWh; x-axis zoomed to the corresponding cumulative-quantity range.

Reading (mixed). CCGT and Hydro carry the bid-shape variation across hours: Iberdrola, Naturgy and EDP-Spain show clear spread of red curves on those columns. Within-hour quarter dispersion is concentrated in Naturgy CCGT and Endesa Hydro — the rest of the (firm, tech) cells produce overlapping Q1–Q4 curves. Nuclear, Wind and Solar PV are flat across hours and quarters, as expected of must-run and price-taker technologies. Empty cells reflect firms with no units of that tech under the primary-owner allocation (Endesa wind via Enel Green Power is registered separately; Naturgy and EDP-Spain hold no nuclear).

A.12 Per-firm demand-side bid curves, all technologies (DA)

Buy bids only (downward-sloping demand curves). Columns: `Pump_load` (pumped-storage charge), `Retailer`, `Direct_consumer`.

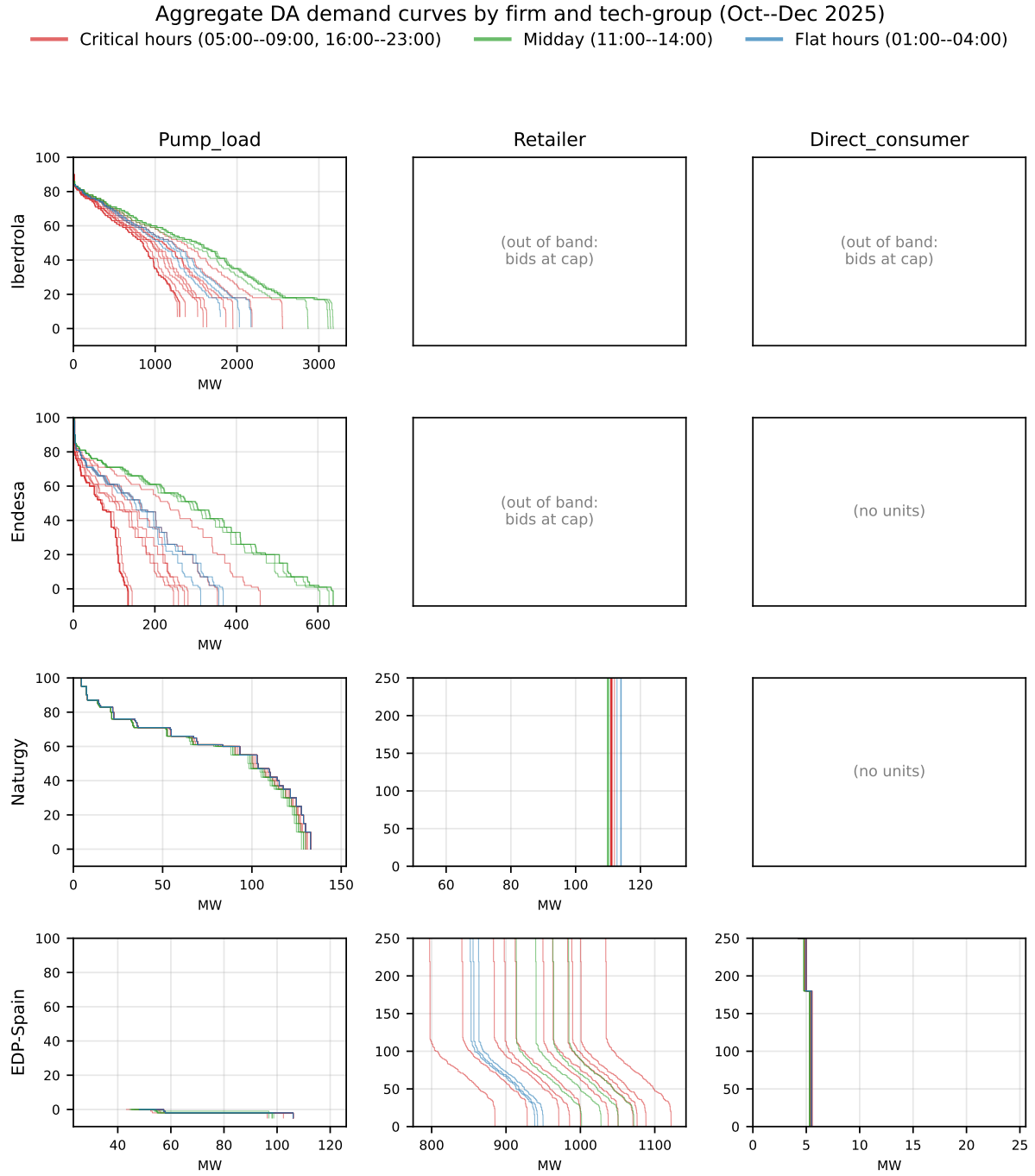


Figure 16: Aggregate DA demand curves by firm and demand-side tech-group, October–December 2025, by hour-class. Rows: pivotal firms. Columns: buy-side tech-groups. Y-band fixed per tech to the strategically informative range; empty panels indicate firms with no units of that tech (or, for Retailer at IB/GE/Naturgy, bids placed exclusively at the price cap so no activity falls inside the band).

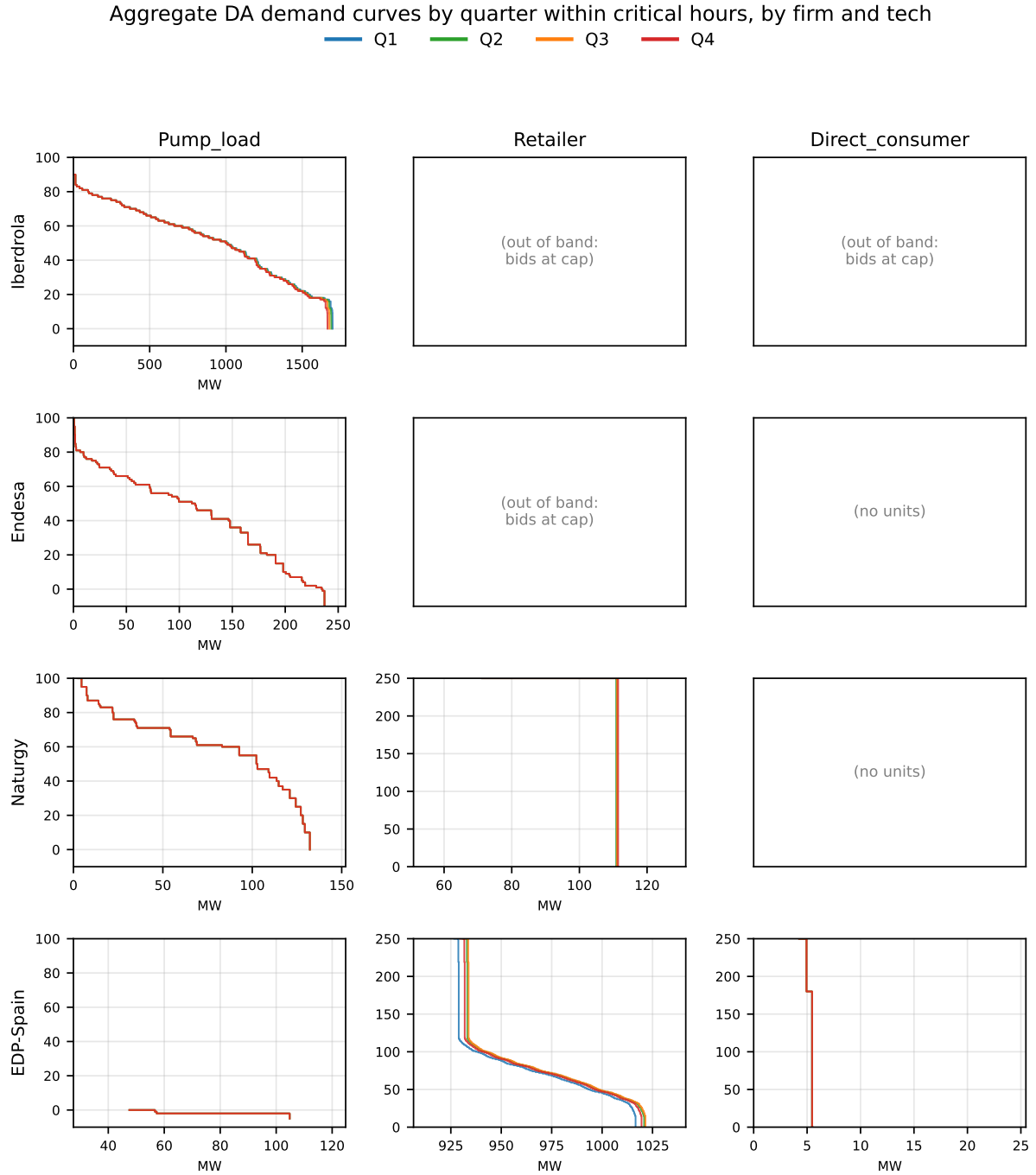


Figure 17: Aggregate DA demand curves by quarter within critical hours, October–December 2025, by firm and demand-side tech-group.

Reading (mixed). At the clock-hour level, **Pump_load** shows clear hour-class discrimination for IB, GE and GN (midday-green curves shifted right of flat-blue, both right of critical-red). The **Retailer** column is empty for IB, GE and Naturgy because their retail buy bids are placed entirely at the price cap — they treat retail as a quantity-securing instrument, not a price signal; only EDP-Spain bids strategically. **Direct_consumer** books are small and concentrated near the cap. At the quarter-within-hour level, every (firm $\times$ tech) cell collapses to a single curve: Q1–Q4 overlap. Within-hour granularity is unused on the demand side.

A.13 Per-firm storage-discharge supply curves (DA)

Sell bids only (upward-sloping supply curves). Columns: **Hydro_pump** (pumped-storage discharge) and **Hybrid_RES_storage**.

Aggregate DA storage discharge curves by firm and tech-group (Oct--Dec 2025)
— Critical hours (05:00--09:00, 16:00--23:00) — Midday (11:00--14:00) — Flat hours (01:00--04:00)

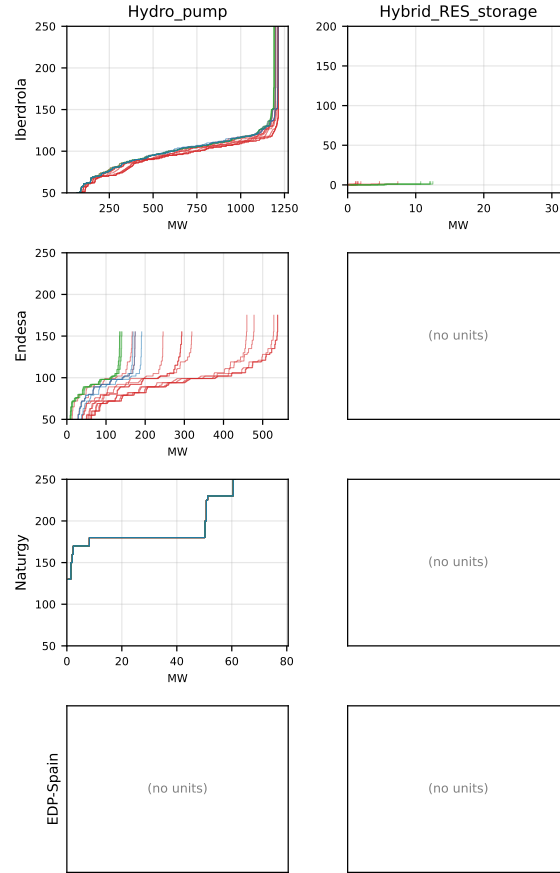


Figure 18: Aggregate DA storage-discharge supply curves by firm and tech-group, October–December 2025, by hour-class. Rows: pivotal firms. Columns: sell-side storage tech-groups.

Aggregate DA storage discharge curves by quarter within critical hours, by firm and tech

— Q1 — Q2 — Q3 — Q4

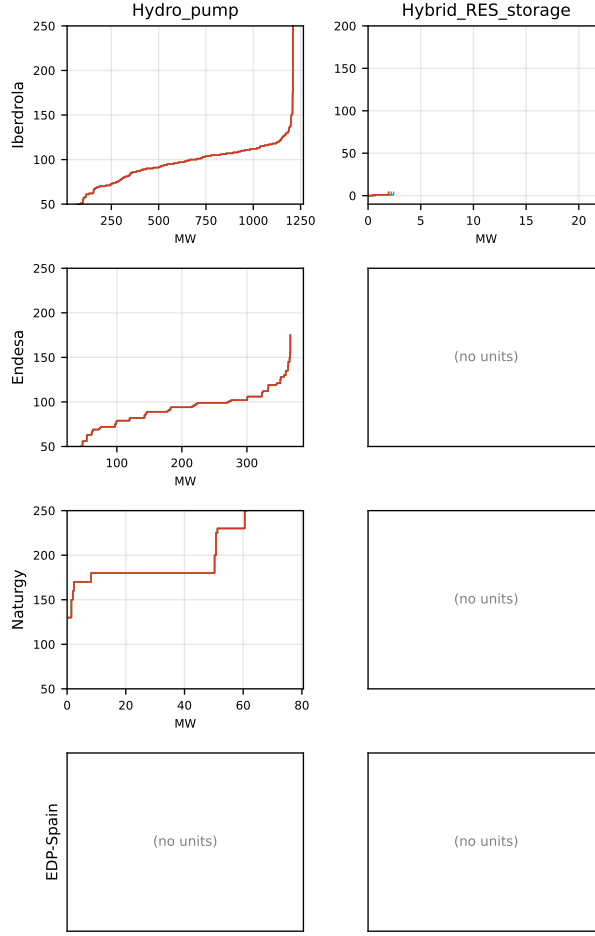


Figure 19: Aggregate DA storage-discharge supply curves by quarter within critical hours, October–December 2025, by firm and tech-group.

Reading (mixed). Hydro_pump discharge curves separate by hour-class for Iberdrola and Endesa (consistent with discharging more in critical hours when prices are highest); Naturgy is small and concentrated; EDP-Spain has no Hydro_pump. Hybrid_RES_storage sell-side books are very small (IB-only with meaningful volume) and bid near zero EUR/MWh. As on the demand side, within-hour quarter dispersion is absent across all (firm $\times$ tech) cells.

A.14 Per-firm demand-side bid curves, all technologies (IDA)

IDA mirror of Appendix A.12. Same construction on ICAB+IDET data, pooled across the 3 IDA sessions per day.

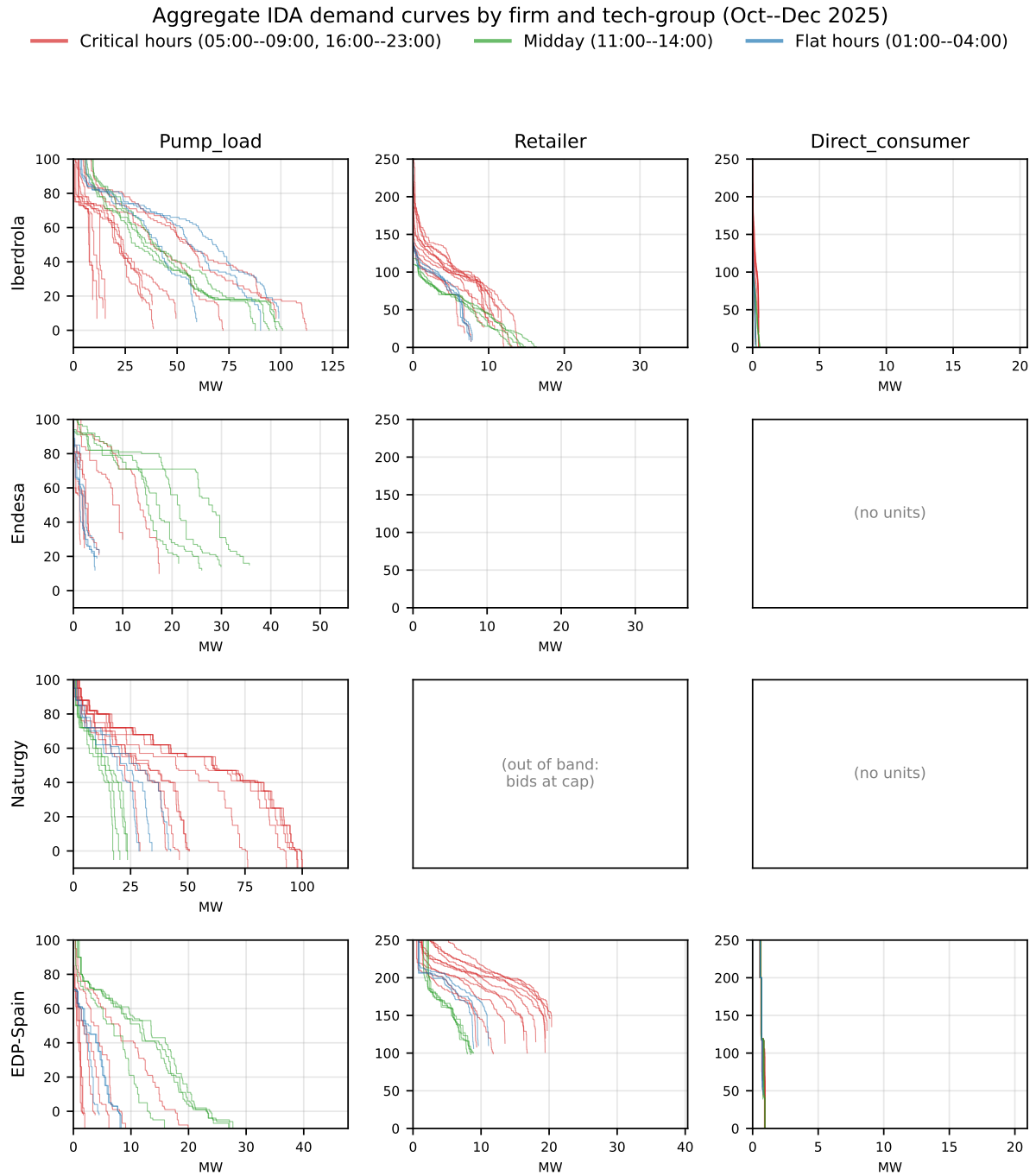


Figure 20: Aggregate IDA demand curves by firm and demand-side tech-group, October–December 2025, by hour-class.

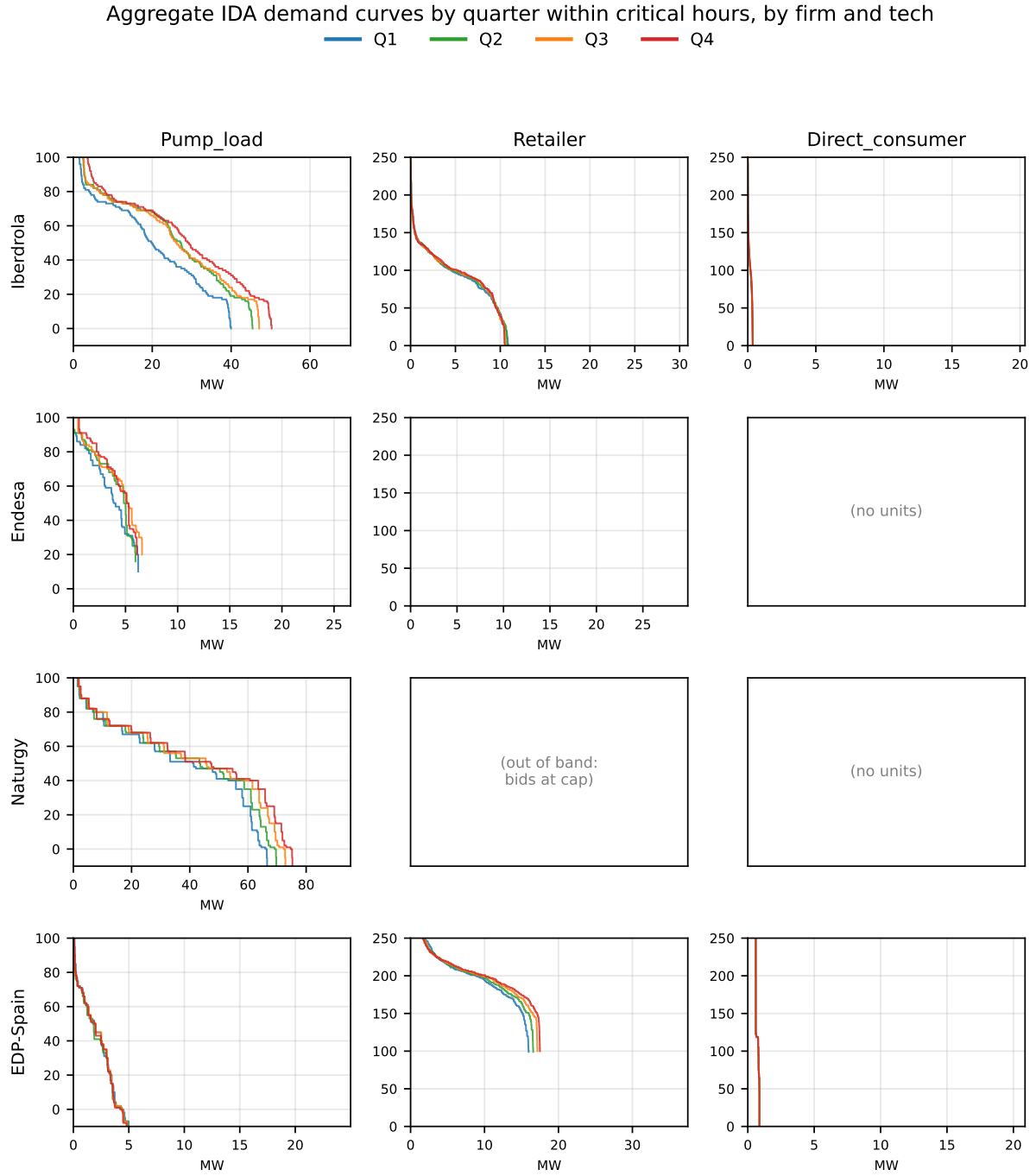


Figure 21: Aggregate IDA demand curves by quarter within critical hours, October–December 2025, by firm and demand-side tech-group.

A.15 Per-firm storage-discharge supply curves (IDA)

IDA mirror of Appendix A.13.

Aggregate IDA storage discharge curves by firm and tech-group (Oct--Dec 2025)
 — Critical hours (05:00--09:00, 16:00--23:00) — Midday (11:00--14:00) — Flat hours (01:00--04:00)

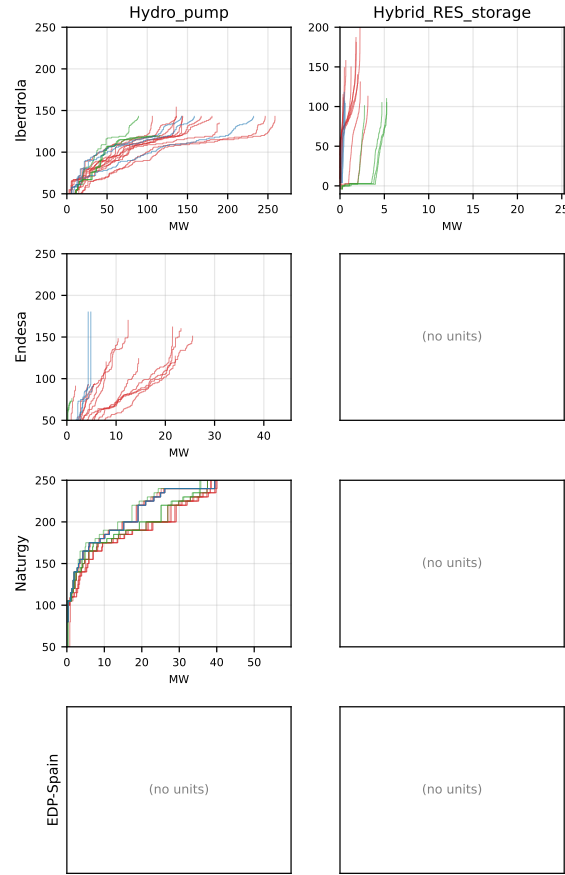


Figure 22: Aggregate IDA storage-discharge supply curves by firm and tech-group, October–December 2025, by hour-class.

Aggregate IDA storage discharge curves by quarter within critical hours, by firm and tech

— Q1 — Q2 — Q3 — Q4

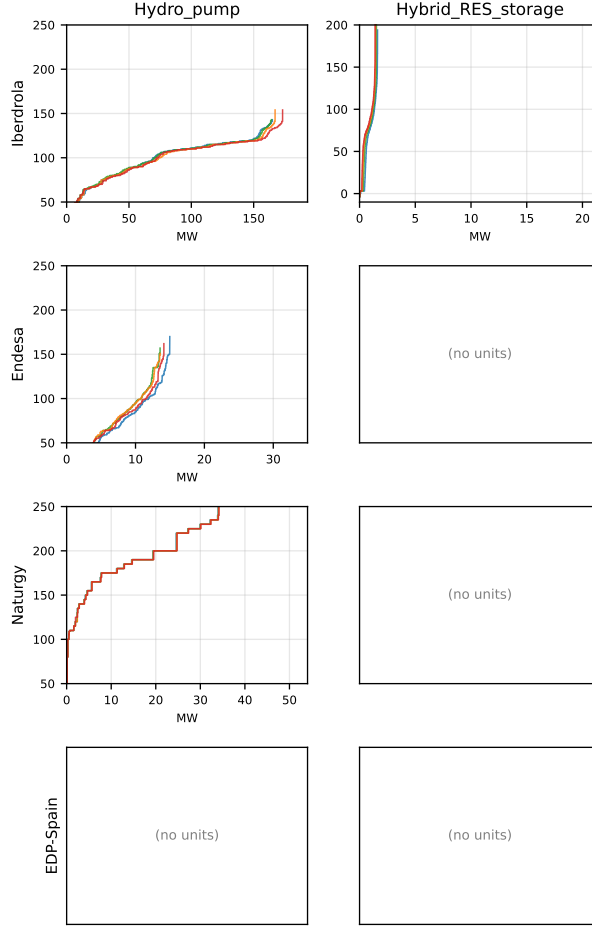


Figure 23: Aggregate IDA storage-discharge supply curves by quarter within critical hours, October–December 2025, by firm and tech-group.

A.16 Within-hour quarter dissimilarity test

Falsification test for the aggregated Q1–Q4 plots: a perfect overlap in the cross-cell average could mask cell-level variation if its direction is idiosyncratic. The diagnostic operates at the (firm, unit, date, hour) cell level, before any aggregation.

Construction. Each quarter’s bid in a cell is a discrete measure on the price axis: each tranche (p_k, q_k) contributes a point mass q_k MWh at price p_k EUR/MWh, after integer-rounding the price. Let μ_i denote the bid measure for quarter $i \in \{1, 2, 3, 4\}$, with cumulative distribution

$$F_i(p) = \sum_{k: p_k \leq p} q_k,$$

which is exactly the bid-curve step function plotted in §4.2: at price p on the y-axis, $F_i(p)$ is the cumulative MWh offered up to that price.

Dissimilarity between two quarters. For an ordered pair (i, j) , we compute the L1 area between the cumulative bid curves over the union of all bid prices present in either quarter:

$$W_1(\mu_i, \mu_j) = \int_{p_{\min}}^{p_{\max}} |F_i(p) - F_j(p)| dp = \sum_{k=1}^{K-1} |F_i(p_{(k)}) - F_j(p_{(k)})| \cdot (p_{(k+1)} - p_{(k)}), \quad (4)$$

where $\{p_{(1)} < p_{(2)} < \dots < p_{(K)}\}$ is the sorted union of distinct bid prices from both quarters and $[p_{\min}, p_{\max}] = [p_{(1)}, p_{(K)}]$ is the resulting integration domain. Computation handles the union grid in $O(K \log K)$ via a sort/merge of the two ordered price lists; see `scripts/analysis/bid/quarter_dissimilarity.py`.

Wasserstein interpretation. Equation 4 is the closed-form expression of the 1-D Wasserstein-1 distance (Earth Mover’s Distance) for probability measures on $\mathbb{R}$: $W_1(\mu, \nu) = \int |F_\mu - F_\nu|$ (Vallender, 1973). Strictly, the equality requires the two measures to share total mass. In our setting, two quarters may submit slightly different total volumes ($\sum_k q_{i,k} \neq \sum_k q_{j,k}$), in which case Equation 4 is the natural *unbalanced* extension — the L1 area between two non-decreasing step functions over the observed price range. This still captures (i) horizontal shifts of the curve (price re-pricing), (ii) vertical shifts (quantity re-allocation), and (iii) total-volume changes, in a single non-negative number. Magnitudes (in MWh·EUR/MWh) are not directly interpretable, but the metric serves the role we need: $W_1 = 0$ **iff the two bid multisets are identical**, and $W_1 > 0$ iff there is any difference.

Per-cell statistic and flag. For each cell with all four quarters present we compute the maximum pairwise dissimilarity:

$$D = \max_{1 \leq i < j \leq 4} W_1(\mu_i, \mu_j).$$

A cell is “flagged” if $D > 10^{-6}$ MWh·EUR/MWh — equivalent to “the four quarters are not bit-identical”.

Table 8: Quarter-by-quarter bid dissimilarity, October–December 2025. For each (firm, unit, date, hour) cell with all four 15-min quarters present, D is the maximum pairwise L1 area between cumulative bid curves. Flag = fraction of cells with $D > 0$.

Side	Tech	Firm	Hour-class	N cells	% flagged	median D
Supply	CCGT	IB	Critical	7,831	5.8	0.00
		IB	Flat	2,139	3.9	0.00
		GE	Critical	5,195	1.8	0.00
		GE	Flat	1,413	0.0	0.00
		GN	Critical	14,930	24.5	0.00
		GN	Flat	4,065	0.0	0.00
		HC	Critical	2,013	0.0	0.00
		HC	Flat	549	0.5	0.00
	Hydro	IB	Critical	6,072	2.1	0.00
		IB	Flat	1,656	0.3	0.00
		GE	Critical	5,552	56.4	0.80
		GE	Flat	1,488	47.3	0.00
		GN	Critical	2,993	0.0	0.00
		GN	Flat	810	0.1	0.00
		HC	Critical	847	0.0	0.00
		HC	Flat	231	0.0	0.00
	Hydro-pump	IB	Critical	1,001	0.4	0.00
		IB	Flat	273	0.0	0.00

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Side	Tech	Firm	Hour-class	N cells	% flagged	median D
	Nuclear	GE	Critical	2,670	0.0	0.00
		GE	Flat	372	0.0	0.00
		GN	Critical	1,006	0.0	0.00
		GN	Flat	276	0.0	0.00
		IB	Critical	2,907	31.9	0.00
		IB	Flat	800	31.6	0.00
		GE	Critical	2,473	4.0	0.00
		GE	Flat	672	1.9	0.00
	Wind	IB	Critical	4,539	97.2	1.00
		IB	Flat	1,283	97.3	1.00
		GN	Critical	4,152	92.7	0.13
		GN	Flat	1,159	92.6	0.10
		HC	Critical	3,990	93.7	2.60
		HC	Flat	1,081	93.8	2.60
	Solar PV	IB	Critical	960	99.5	7.80
		GE	Critical	64	3.1	0.00
		GN	Critical	1,594	15.9	0.00
		GN	Flat	411	0.0	0.00
		HC	Critical	325	100.0	42.12
Demand	Pump_load	IB	Critical	2,558	3.2	0.00
		IB	Flat	866	5.5	0.00
		GE	Critical	1,370	0.0	0.00
		GE	Flat	617	0.0	0.00
		GN	Critical	1,010	0.0	0.00
		GN	Flat	276	0.0	0.00
		HC	Critical	946	0.0	0.00
		HC	Flat	261	0.0	0.00
	Retailer	IB	Critical	3,423	40.9	0.00
		IB	Flat	923	40.2	0.00
		GE	Critical	3,971	76.5	11.20
		GE	Flat	1,083	76.3	8.90
		GN	Critical	4,048	22.0	0.00
		GN	Flat	1,104	15.8	0.00
		HC	Critical	1,100	0.5	0.00
		HC	Flat	300	0.0	0.00
	Direct_consumer	IB	Critical	1,012	95.9	0.30
		IB	Flat	276	95.3	0.30
		HC	Critical	1,612	15.6	0.00
		HC	Flat	441	12.2	0.00

Reading (mixed). Pump-storage is genuinely mechanical: **Pump_load** and **Hydro_pump** flag $\leq 5.5\%$ across all firms and hour-classes, confirming the within-hour null in §4.3. Naturgy CCGT flags 24.5% in critical vs 0.0% in flat, matching the visual Q1–Q4 separation in Figure 4a. By contrast **Wind** and **Solar PV** flag $\geq 92\%$ for IB, GN and HC — bids vary across the four quarters in almost every cell, though for Wind the magnitudes are small (median $D \leq 2.6$), consistent with quarter-by-quarter automated forecast updates rather than strategic re-pricing. **Retailer** flags 41%, 76% and 22% for IB, GE and GN with Endesa’s median $D = 11.2$: pivotal firms do vary their retail demand bids across quarters in ways the aggregated demand-curve plot of §A.12 does not display. Endesa Hydro stands out on the supply side with 56% / 47% flag rates.

A.17 Pre-trend diagnostic

The parallel-trends assumption requires $\mathbb{E}[Y_t(0) - Y_0(0) \mid \text{crit}] = \mathbb{E}[Y_t(0) - Y_0(0) \mid \text{flat}]$ in the pre-reform window: the time-evolution of the counterfactual outcome should be the same for critical and flat hours. The visual diagnostic below plots, for each outcome, the cumulative deviation of the monthly mean from its January-2024 baseline in critical hours (red) and flat hours (blue). Two lines that overlap pre-reform support the assumption.

A.17.1 Outcome 1: q_2 (intraday upward repositioning)

OJO, MOEVE SHOWS A HUGE CHANGE IN BEHAVIOUR AFTER THE ID15 REFORM!!
THE REST SEEM NICE!!

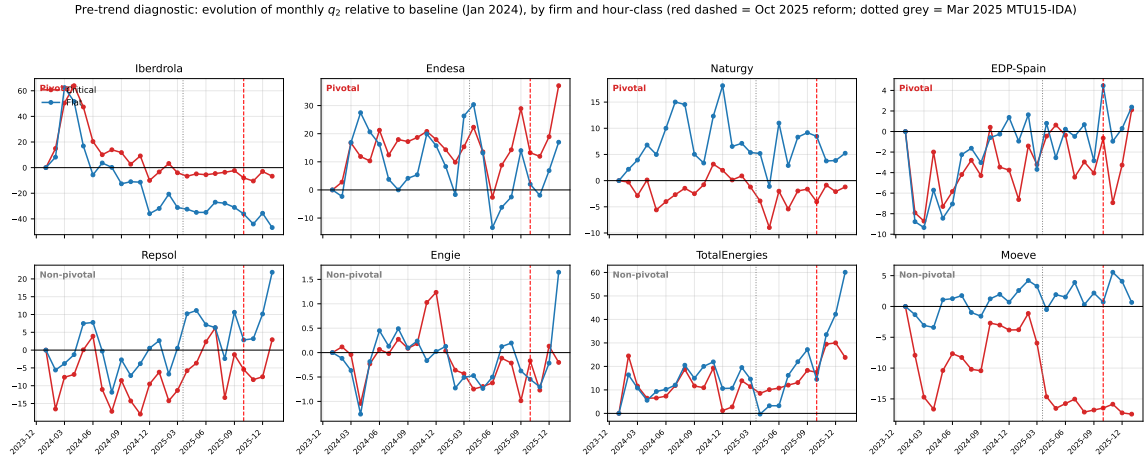


Figure 24: Per-firm evolution of monthly q_2 relative to Jan-2024 baseline, by hour-class, Jan 2024 – Dec 2025. Each panel plots, for one firm, the cumulative deviation of monthly mean q_2 from its January-2024 value in critical hours (red) and flat hours (blue). Parallel-trends test: the two lines should track each other pre-reform. Red dashed: MTU15-DA reform. Grey dotted: MTU15-IDA. EDP-PT omitted (no PIBCI entries).

OJO, ID15 IS VERY IMPORTANT FOR WIND!! I SHOULD THINK ABOUT HOW IT AFFECTS!!

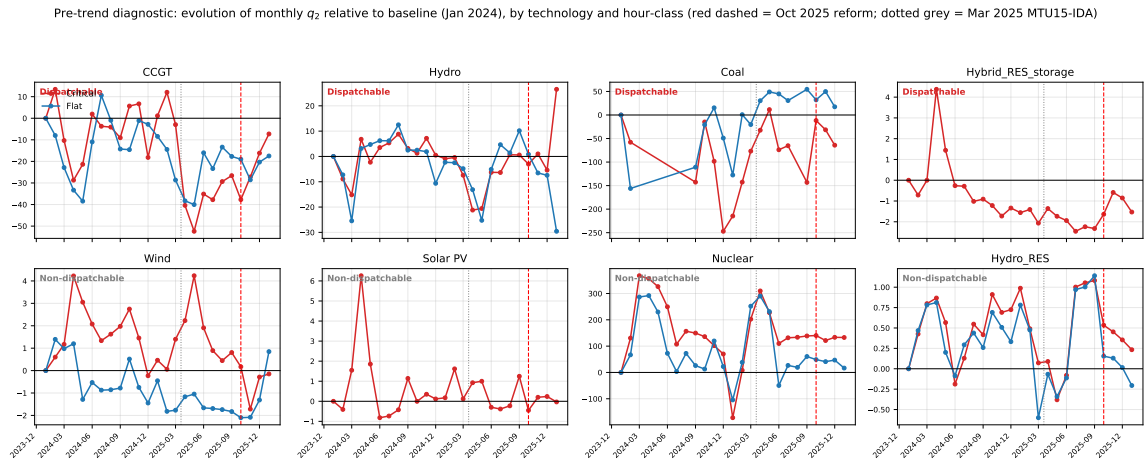


Figure 25: Per-technology evolution of monthly q_2 relative to Jan-2024 baseline, by hour-class, Jan 2024 – Dec 2025. Same construction as Figure 24 but grouped by tech_group instead of parent. Sample restricted to pivotal + non-pivotal firms. Solar PV has no flat-hour observations by construction (no generation at 1–3 AM); Coal sample is small.

Reading (mixed). For most firms the red and blue lines track each other pre-reform, suggesting parallel trends are plausible at the firm level. The exception is Moeve, whose flat-hour series shifts down sharply at MTU15-IDA (Mar 2025) and stays below baseline thereafter, while the critical-hour series stays near zero — a structural break in flat-hour behaviour that is concentrated in this single non-pivotal firm. By technology, CCGT and Hydro pre-trends look reasonable; Wind, Nuclear and Hydro-RES also co-move pre-reform. The headline $\beta_3 \approx +3.87$ sits inside the month-to-month noise band, so the DiD identification leans on averaging across day-clusters within the same-cal-month window rather than on any single month.

A.17.2 Outcome 2: clearing prices (IDA and DA)

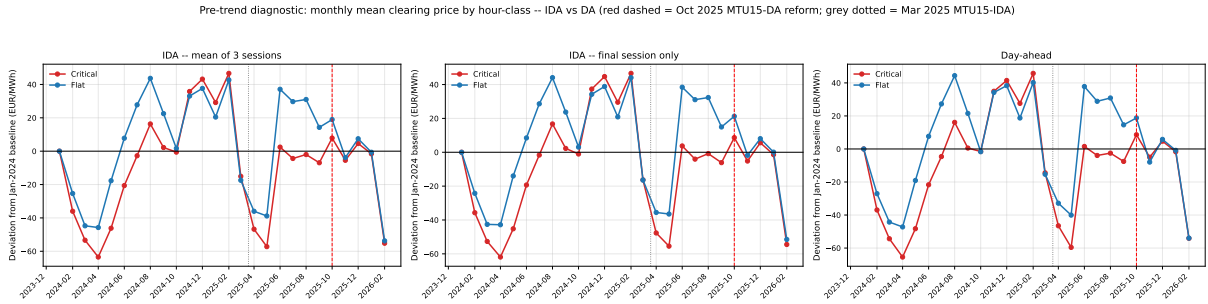


Figure 26: Evolution of monthly mean Spanish clearing price relative to Jan-2024 baseline, by hour-class, Jan 2024 – Dec 2025. Left: arithmetic mean across the three IDA auction sessions covering each (date, clock-hour). Middle: only the last (final) IDA session covering each slot. Right: day-ahead clearing price. All three average across the MTU15 periods that map to the same clock-hour before averaging within (month, hour-class). Red dashed: MTU15-DA reform. Grey dotted: MTU15-IDA.

Reading (supportive). All three markets tell the same story. Pre-MTU15-IDA, the critical and flat lines track each other tightly through the seasonal cycle (down to -60 EUR/MWh in spring 2024, up to $+45$ by Feb 2025): the crit-flat price spread is stable, parallel trends look strong. After MTU15-IDA (Mar 2025) the lines diverge: flat-hour prices rise $+30$ to $+40$ above baseline while critical-hour prices stay near zero, compressing the peak-flat spread by 30 – 40 EUR/MWh on average. This is consistent with the strategic-granularity story — the reform lets dispatchable units re-price into critical hours, pulling those clearing prices down relative to flat hours where there is no analogous opportunity. The fact that the pattern is essentially identical in DA and IDA reflects the arbitrage-linked nature of the sequential markets: the DA-to-IDA wedge is small (Appendix A.6). The mean-of-3-sessions and final-session-only IDA variants are visually indistinguishable.

A.18 Time placebos

A.19 Triple-difference: pivotality as moderator

$$\begin{aligned}
 q_{2,fudh} = & \alpha + \beta_1 \text{crit}_h + \beta_2 \text{post}_d + \beta_3 \text{piv}_f \\
 & + \beta_{12} \text{crit}_h \text{post}_d + \beta_{13} \text{crit}_h \text{piv}_f + \beta_{23} \text{post}_d \text{piv}_f \\
 & + \beta_{123} \text{crit}_h \text{post}_d \text{piv}_f + \gamma_f + \delta_{\text{DOW}(d)} + \theta' X_{dh} + \varepsilon_{fudh}, \quad (5)
 \end{aligned}$$

where $X_{dh} = (\text{wind}_{dh}, \text{solar}_{dh}, \text{demand}_{dh})$ in GW.

B Data appendix

B.1 Sequential-market technical reference (PDBC, PDBF, PDVD, PIBCA, PHF)

B.2 Firms in the analysis sample (detailed)

I THINK WE CAN TAKE THE CAPACITY DATA! BUT MAYBE IT IS ONLY FOR EXPORTS AND IMPORTS!.

Column definitions. *DA auction share*: PDBF offer-type 1, anonymous DA clearing. *DA bilateral share*: PDBF offer-type 4, pre-negotiated firm-to-firm transfers. *IDA cumulative share*: PIBCA accumulated (last session per unit-period). *Net position*: net seller if 2025 DA sales > 60% of turnover, net buyer if < 40%, mixed otherwise. The net-position label is descriptive; vertical integration does not enter the identification.

Table 9: Firms in the analysis sample (detailed). Channel-segregated 2025 shares. Channel denominators are channel-specific (they do not sum across rows).

Parent firm	Gen. units	Capacity (GW)	DA auction share (%)	DA bilateral share (%)	IDA cumul. share (%)	Net position
<i>Dominant operators in Spanish generation (CNMC tier)</i>						
Iberdrola	70	31.03	20.01	27.18	20.37	Mixed
Endesa	38	12.46	36.88	7.11	12.64	Net buyer
Naturgy	39	12.60	3.86	16.18	7.67	Net buyer
EDP-Spain	21	4.73	0.77	5.67	2.60	Net buyer
EDP-Portugal	16	7.42	20.62	12.70	5.17	—
<i>Other large operators (non-dominant)</i>						
Repsol	13	0.61	1.79	1.26	0.62	Net buyer
Engie España	59	4.05	0.03	2.49	1.07	Net seller
TotalEnergies	15	1.19	1.19	0.38	0.59	Net buyer
Moeve	112	1.28	0.03	—	0.66	Net buyer

Reading (descriptive). Iberdrola is roughly balanced across channels. Endesa is auction-heavy. Naturgy is bilateral-heavy. Vertical-integration status is descriptive only and does not enter the identification.

B.3 File-format specifications and parser conventions

B.4 Power-vs-energy unit discipline

B.5 Coverage and missingness

B.6 Hour-of-day classification metrics

Table 10: Hour-of-day classification metrics, Spain 2025. Mean load, solar, net-load (= load – wind – solar), within-hour SD of net-load, and signed within-hour load change $\Delta\text{load} = \text{load}_{q_4} - \text{load}_{q_1}$, by clock-hour.

Clock-hour	Demand (GW)	Solar (GW)	Wind (GW)	Residual demand (GW)	σ_{within} (MW)	Δ demand (MW)	Class
00:00–01:00	23.2	0.2	7.1	16.0	405	–864	<i>dropped</i>
01:00–02:00	22.4	0.1	6.8	15.4	353	–481	<i>flat</i>
02:00–03:00	21.9	0.1	6.6	15.1	269	–193	<i>flat</i>
03:00–04:00	21.9	0.1	6.5	15.3	324	194	<i>flat</i>
04:00–05:00	22.9	0.1	6.4	16.4	649	1179	<i>dropped</i>
05:00–06:00	24.9	0.3	6.3	18.3	766	1795	<i>critical</i>
06:00–07:00	27.0	1.9	6.2	18.9	1009	1420	<i>critical</i>
07:00–08:00	28.3	5.9	5.8	16.7	1259	602	<i>critical</i>
08:00–09:00	28.7	10.5	5.3	12.9	1116	218	<i>critical</i>
09:00–10:00	28.7	13.6	5.0	10.1	783	–103	<i>dropped</i>
10:00–11:00	28.6	15.0	4.8	8.7	576	–78	<i>dropped</i>
11:00–12:00	28.6	15.6	4.9	8.1	471	116	<i>dropped</i>
12:00–13:00	28.7	15.6	5.1	8.0	448	–60	<i>dropped</i>
13:00–14:00	28.3	15.2	5.3	7.9	453	–290	<i>dropped</i>
14:00–15:00	28.0	14.4	5.5	8.1	545	–57	<i>dropped</i>
15:00–16:00	28.3	13.0	5.8	9.4	747	344	<i>dropped</i>
16:00–17:00	28.9	10.1	6.3	12.5	1241	634	<i>critical</i>
17:00–18:00	30.0	6.4	6.8	16.8	1362	902	<i>critical</i>
18:00–19:00	31.2	2.7	7.3	21.2	1185	847	<i>critical</i>
19:00–20:00	32.0	0.7	7.4	23.9	557	163	<i>critical</i>
20:00–21:00	31.1	0.3	7.5	23.2	614	–1487	<i>critical</i>
21:00–22:00	28.6	0.2	7.5	20.9	801	–1926	<i>critical</i>
22:00–23:00	26.4	0.2	7.4	18.8	694	–1441	<i>critical</i>
23:00–00:00	24.6	0.2	7.2	17.2	534	–1093	<i>dropped</i>

Reading (supportive). Critical hours (05:00–09:00, 16:00–23:00) have within-hour SD of net-load several times higher than flat hours (01:00–04:00). The ranking pins the classification directly to the structural variability of residual demand.

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